# Reciprocal Package Coprimality for the Polynomials $Q_{a,b}$

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**Abstract**

For positive integers $a, b$, set

$$Q_{a,b}(x) = \frac{ax^{a+b} - (a+b)x^a + b}{(x^{\gcd(a,b)} - 1)^2}.$$

The two polynomials $Q_{a,b}$ and $Q_{b,a}$ are reciprocal partners, so an unordered pair $\{a, b\}$ naturally carries the package $P_{a,b} = Q_{a,b}Q_{b,a}$. We prove that these packages are pairwise coprime over $\mathbb{Q}$: if $\{a, b\} \neq \{c, d\}$ and both pairs are unequal, then $\gcd(P_{a,b}, P_{c,d}) = 1$ in $\mathbb{Q}[x]$.

The proof converts a hypothetical common factor into an off-unit-circle collision among the divided differences $L_k(x) = (1 - x^k)/k$. Analytic rigidity, local power sieves, height bounds, and endpoint arithmetic then force every such collision into an explicit finite list of arithmetic configurations. The remaining case analysis is computer-assisted but fully deterministic: bundled verification scripts, which any reader can replay, eliminate every configuration on the list, the few terminal cases being settled by modular gcd certificates.

## Contents

# 1 Statement, context, and proof map

The central object of the paper is the family

$$Q_{a,b}(x) = \frac{ax^{a+b} - (a+b)x^a + b}{(x^{\gcd(a,b)} - 1)^2} \qquad (a, b \in \mathbb{Z}_{>0}).$$

When $a \neq b$, write

$$P_{a,b}(x) = Q_{a,b}(x)Q_{b,a}(x).$$

The product depends only on the unordered pair $\{a, b\}$; it is the *reciprocal package* attached to that pair. The denominator in $Q_{a,b}$ removes the forced double cyclotomic factor, and Lemma 2.2 below shows that no root of any $Q_{a,b}$ lies on the unit circle.

**Theorem 1.1** (Main package-coprimality theorem)**.** *Let $a, b, c, d$ be positive integers with $a \neq b$, $c \neq d$, and $\{a, b\} \neq \{c, d\}$. Then*

$$\gcd(P_{a,b}, P_{c,d}) = 1 \qquad \textit{in } \mathbb{Q}[x].$$

*Equivalently, no irreducible factor over $\mathbb{Q}$ occurs in the reciprocal packages of two distinct unordered unequal pairs.*

This statement sits between two nearby problems. The strongest natural target would be irreducibility of every nonconstant $Q_{a,b}$; in the primitive case this includes Borisov's irreducibility problem for the *abc*-polynomials [1]. The package-coprimality theorem is weaker than universal irreducibility, but it is uniform in the whole imprimitive family because

$$Q_{rA,rB}(x) = rQ_{A,B}(x^r).$$

It is also strong enough to give each package a factor-theoretic identity that cannot be shared with any other package.

The proof has five layers.

(1) Sections 2–4 translate package sharing into a graph of equal divided differences and prove the first orientation facts: no unit-circle roots, no three equal values, disjoint additive triples, and finite support for any fixed factor orbit.

(2) Sections 6–10 add analytic and algebraic rigidity. The tail-conjugacy identity, sign law, phase bounds, Schur deformation, and rational-root classification are used later as structural sieves rather than as a direct contradiction.

(3) Sections 11–17 build the arithmetic reduction: prime separation, $S$-unit finiteness, proper-divisibility rigidity, self-power rigidity, local power sieves, resultant information, sparse Wronskians, and explicit height–degree bounds.

(4) Section 18 eliminates the upper range $6\,816\,242 \leq D \leq 175\,394\,637$. Section 19 then records the lower-range state variables and the exact endpoint conditions used by the finite searches.

(5) Sections 20 and 21 finish the proof by deterministic computer-assisted eliminations of, respectively, the unit-coefficient branch and all nonunit branches.

The shortest dependency path to the final contradiction is:

$$\text{collision dictionary} \Longrightarrow \text{structural reductions and local packet sieves}$$
$$\Longrightarrow \text{upper-range computation} \Longrightarrow \text{lower-state envelope}$$
$$\Longrightarrow \text{unit branch and nonunit branches} \Longrightarrow \text{terminal modular certificates} \Longrightarrow \text{Theorem 1.1.}$$

The computational part is used only after the preceding sections have reduced a counterexample to explicit integer data. Each search is conservative: rows are kept unless a stated mathematical filter removes them, and the verification scripts recompute the arithmetic conditions and terminal finite-field certificates from the recorded data. Thus the trust boundary is the usual one for a reproducible computer-assisted proof package: exact source code, deterministic data files, checksums, and independent replayers, not a formal proof-assistant certificate. Table 1 gives the paper-to-code correspondence: the generator and the independent re-checker behind each computational result.

For reference, the following roadmap lists the main mathematical and computational outputs in the order they are used; it is an index to the proof, not a substitute for the later arguments. Each symbol is defined where it first arises; in brief, $\{a, b\}$ and $\{c, d\}$ are the two packages, a counterexample is a shared irreducible factor whose normalization yields integers $0 < m < n$ and $0 < p < q$ with $\gcd(m, n, p, q) = 1$ and $D = \max\{m, n, p, q\}$; the integers $r, s$ are the scales of the two primitive shapes; $d$ is the degree over $\mathbb{Q}$ of a common root; $e$ a factor degree; $U$ and $V$ the leading and constant coefficients of the primitive minimal polynomial of a common root; and $M(\cdot)$ the Mahler measure.

(i) Every noncyclotomic irreducible reciprocal factor orbit occurs in only finitely many packages. Hence the package-sharing graph is locally finite.

(ii) For every primitive unequal pair $(a, b)$, including boundary pairs,

$$Q_{a,b} \mid Q_{c,d} \quad \Longrightarrow \quad (c, d) = (a, b).$$

(iii) Let $A, B$ be coprime and unequal, and suppose $Q_{A,B}$ is irreducible. If $\beta$ is a root, then

$$\beta \notin \mathbb{Q}(\beta)^p \qquad \text{for every prime } p.$$

Every lift $\{rA, rB\}$ is therefore globally isolated.

(iv) If $I(X)$ counts unordered unequal pairs of total at most $X$ whose packages are not globally isolated, then

$$I(X) = O\Big(X^{20/11} \log X\Big).$$

Thus a density-one set of package vertices is globally isolated from the entire infinite family.

(v) If an irreducible factor of degree $e$ divides a primitive $Q_{A,B}$, then

$$\operatorname{rad}(A + B) \mid e(e + 2).$$

For every prime $p \mid A + B$, including $p = 2$ with arbitrary $v_2(A + B)$, a factor root is not a $p$-th power in its own field.

(vi) If $p^\kappa \| B$ and a degree-$e$ factor root is a $p$-th power in its own field, then

$$e \le A, \qquad pA \mid e\kappa,$$

for every prime $p$, with the reciprocal statement when $p^\kappa \| A$. Thus the only local power room at a bad endpoint is inside its nonunit block, and it requires $\kappa \ge p$.

(vii) If $N = qk$ with $q$ odd prime and $Q_{A,B} = \prod_i h_i$, then the factor degrees satisfy the exact partition identities of Corollary 15.11. In particular, if $q > k+1$, exactly one factor has degree congruent to $-2 \pmod q$ and there are at most $k$ factors. As a consequence, primitive total $q$ or $2q$, with $q$ an odd prime, always gives an irreducible $Q_{A,B}$.

(viii) Every normalized four-index counterexample satisfies

$$\max\{m, n, p, q\} \le 175\,394\,637.$$

If the maximum is at least $6\,816\,242$, the primitive minimal polynomial of a common root is monic with constant coefficient $\pm 1$.

(ix) An exact integer computation eliminates that entire unit-normalized range. Consequently every normalized counterexample satisfies

$$\boxed{\max\{m, n, p, q\} \le 6\,816\,241.}$$

The computation is reduced to 761 same-shape rows, all excluded by the incompatible bounds $d \le 2rs \le 38\,364$ and $d \ge 48\,688$.

(x) A second exact computation eliminates every counterexample in the remaining box whose primitive minimal polynomial has unit leading and constant coefficients. Therefore every counterexample is genuinely nonunit.

(xi) The nonunit computations eliminate the both-nonunit branch, the one-nonunit branch with $D \ge 16\,584$, and the residual one-nonunit branch

$$D \le 16\,583, \qquad |V| = 1, \qquad 2 \le U \le D, \qquad d \le 2601.$$

The residual branch produces 233 278 conservative package states, four terminal state pairs, two package pairs, and eight modular gcd certificates, all replayed by the bundled verifier. Consequently no normalized counterexample remains.

(xii) If a primitive irreducible polynomial $h$, of degree $e$ and Mahler measure $M(h)$, divides selected orientations of two primitive packages, then the radical of the product of their six additive-triple entries is at most

$$2eM(h)^{4-1/e},$$

and at most $2eM(h)^{2-1/e}$ in the monic, unit-constant case. The leading and constant coefficients also satisfy the four endpoint divisibilities (17.24), and their prime divisors force the exact slope matching (17.25)–(17.26).

Items (viii)–(xi) are the completion path for Theorem 1.1. The height–degree argument and the computation in Section 18 eliminate the upper unit-normalized range; Section 20 eliminates all unit factors in the remaining box; and Section 21 eliminates the both-nonunit branch, the one-nonunit branch with $D \ge 16\,584$, and the residual one-nonunit range $D \le 16\,583$. Thus no residual branch remains after the finite arithmetic eliminations.

## 2 Reciprocal normalization and the collision dictionary

The first task is to replace the factor question by an object that can be followed combinatorially. This section introduces the divided differences $L_k$, removes the cyclotomic double roots, and records the primitive normalization used throughout the rest of the proof.

For $0 < m < n$, define

$$H_{m,n}(x) = mx^n - nx^m + (n-m), \qquad L_k(x) = \frac{1-x^k}{k}.$$

Then

$$H_{m,n}(z) = 0 \quad \Longleftrightarrow \quad L_m(z) = L_n(z), \tag{2.1}$$

and

$$Q_{a,b}(x) = \frac{H_{a,a+b}(x)}{(x^{\gcd(a,b)}-1)^2}. \tag{2.2}$$

### 2.1 Reciprocal packages

Let $g = \gcd(a,b)$ and $N = a+b$. Direct calculation gives

$$x^{N-2g} Q_{a,b}(1/x) = Q_{b,a}(x). \tag{2.3}$$

The monomial in (2.3) is harmless because $Q_{a,b}(0) = b \neq 0$.

**Definition 2.1.** The reciprocal package of the unordered pair $\{a,b\}$ is

$$\mathcal{P}_{\{a,b\}} = \{Q_{a,b}, Q_{b,a}\},$$

or, equivalently for factor questions, the product $Q_{a,b}Q_{b,a}$.

An edge $\{m,n\}$, $0 < m < n$, in the collision graph corresponds to the unordered package $\{m, n-m\}$. The two orientations of that package are the complementary edges

$$\{m,n\}, \qquad \{n-m,n\}.$$

### 2.2 The unit circle is completely removed

We first show that dividing out the double cyclotomic factor removes every unit-circle root, so that all collisions of interest occur off the unit circle.

**Lemma 2.2** (Unit-circle roots)**.** *Let $e = \gcd(m,n)$. If $H_{m,n}(z) = 0$ and $|z| = 1$, then*

$$z^m = z^n = 1, \qquad \textit{hence} \qquad z^e = 1.$$

*Conversely, every $e$-th root of unity is an exactly double root of $H_{m,n}$. Consequently,*

$$\frac{H_{m,n}(x)}{(x^e-1)^2}$$

*has no root on the unit circle.*

*Proof.* The collision equation can be written

$$z^m = \left(1 - \frac{m}{n}\right) + \frac{m}{n} z^n.$$

The right side is a strict convex combination of two points of the closed unit disk. Its modulus is one, so equality in the triangle inequality forces $z^n = 1$, and then $z^m = 1$.

If $\zeta^e = 1$, then $H_{m,n}(\zeta) = H'_{m,n}(\zeta) = 0$, while

$$H''_{m,n}(\zeta) = mn(n-m)\zeta^{m-2} \neq 0.$$

Thus the multiplicity is exactly two. These are all the unit-circle roots by the first paragraph. □

These are also all multiple roots. Indeed,

$$H'_{m,n}(x) = mnx^{m-1}(x^{n-m} - 1).$$

If $H_{m,n}(x) = H'_{m,n}(x) = 0$, then $x \neq 0$, $x^{n-m} = 1$, and therefore

$$0 = H_{m,n}(x) = (n-m)(1 - x^m).$$

Thus $x^m = x^n = 1$, so $x^{\gcd(m,n)} = 1$. Consequently every $Q_{a,b}$ is square-free and all of its roots lie off the unit circle. Also

$$H_{m,2m}(x) = m(x^m - 1)^2, \qquad Q_{m,m}(x) = m,$$

so an off-unit-circle collision always corresponds to an unequal pair.

### 2.3 Primitive reduction

If $a = gA$, $b = gB$, and $\gcd(A, B) = 1$, then

$$Q_{a,b}(x) = gQ_{A,B}(x^g). \tag{2.4}$$

Likewise, if a common-root configuration contains the edges $\{m, n\}$ and $\{p, q\}$, and

$$s = \gcd(m, n, p, q), \qquad w = z^s,$$

then

$$L_{sk}(z) = \frac{1}{s} L_k(w). \tag{2.5}$$

Thus a minimal counterexample may be normalized by

$$\gcd(m, n, p, q) = 1.$$

In that normalization, $\gcd(\gcd(m, n), \gcd(p, q)) = 1$, exactly the coprime-gcd regime occurring in Schinzel's bivariate theorem.

## 3 No three equal values and additive-triple separation

The dictionary from Section 2 is useful only if a single root cannot create many unrelated collisions. The no-three theorem supplies that control: it turns the collision graph at an off-unit-circle point into a matching and forces the additive triples of two hypothetical packages to be disjoint.

For fixed $z$, let $\mathcal{G}_z$ be the graph on the positive integers in which $\{m, n\}$ is an edge precisely when $L_m(z) = L_n(z)$.

**Theorem 3.1** (No-three theorem). *If $|z| \neq 1$, there are no three distinct positive integers $i, j, k$ with*

$$L_i(z) = L_j(z) = L_k(z).$$

*Consequently, $\mathcal{G}_z$ is a matching.*

*Proof.* Let the common value be $A$, and put $R = |z|^2 \neq 1$. Then

$$z^r = 1 - rA \qquad (r = i, j, k).$$

Consider the real smooth function

$$F(t) = |1 - tA|^2 - R^t.$$

It vanishes at the four distinct real points $0, i, j, k$, whereas

$$F'''(t) = -(\log R)^3 R^t$$

never vanishes. Three applications of Rolle's theorem give a contradiction. □

Because the two complementary edges in (2.1) share their upper endpoint, Theorem 3.1 rules out their simultaneous occurrence at an off-unit-circle root. With this in hand, we can state the exact equivalence between package coprimality and the collision graph.

**Proposition 3.2** (Exact collision reformulation). *The reciprocal packages of distinct unordered pairs are pairwise coprime if and only if*

$$|E(\mathcal{G}_z)| \leq 1 \qquad \textit{for every } z \in \mathbb{C}^\times,\ |z| \neq 1.$$

*Proof.* If two package products share an irreducible factor, choose a complex root $z$ of that factor. Lemma 2.2 gives $|z| \neq 1$, and an orientation from each package gives two collision edges at $z$. Conversely, an edge equation has rational coefficients, so a common root of two edges is algebraic; because it is off the unit circle, its minimal polynomial divides the corresponding two $Q$-polynomials. Finally, two edges representing the same unordered package would be the complementary edges in (2.1), which share their upper endpoint and are forbidden by Theorem 3.1. Thus two distinct graph edges are exactly a common factor between two distinct packages. □

The addition law

$$(u + v)L_{u+v}(z) = uL_u(z) + vz^u L_v(z) \tag{3.1}$$

is fundamental.

**Lemma 3.3** (Propagation)**.** *If*

$$L_u = L_{u+v} \qquad \text{and} \qquad L_v = L_w,$$

*then*

$$L_u = L_{u+w}.$$

*If $w \neq v$, this contradicts Theorem 3.1.*

*Proof.* The first equality and (3.1) give $L_u = z^u L_v$. Hence

$$(u+w)L_{u+w} = uL_u + wz^u L_w = uL_u + wz^u L_v = (u+w)L_u.$$

□

For an unordered pair define its additive triple

$$\mathcal{T}(a,b) = \{a, b, a+b\}.$$

**Corollary 3.4** (Disjoint additive triples)**.** *If two distinct packages occur in a hypothetical common-root collision, then*

$$\mathcal{T}(a,b) \cap \mathcal{T}(c,d) = \varnothing. \tag{3.2}$$

*Thus all six integers*

$$a, b, a+b, c, d, c+d$$

*are pairwise distinct.*

*Proof.* Choose one collision edge from each package. Every member of an additive triple is either an endpoint or the gap of its selected edge. A shared endpoint contradicts the matching conclusion of Theorem 3.1. A shared gap becomes a shared endpoint after replacing $z$ by $1/z$. For the endpoint–gap case, write the second edge as $\{r, r+s\}$ and suppose, for example, that $s$ is an endpoint of the first edge $\{s, w\}$. Then

$$L_r = L_{r+s}, \qquad L_s = L_w,$$

and Lemma 3.3 gives $L_r = L_{r+w}$, producing three distinct equal values. The other cross cases are symmetric. These exhaust the possibilities. □

This simultaneously eliminates equal totals, equal gaps, shared entries, and all endpoint–gap cross equalities.

## 4 Finite support of every factor orbit and local finiteness

Before looking for a uniform contradiction, we first prove a qualitative finiteness statement. A fixed noncyclotomic factor orbit can occur in only finitely many packages; this explains why package privacy is, in principle, a finite exact question for each individual package.

For a primitive irreducible polynomial $h \in \mathbb{Z}[x]$ with $h(0) \neq 0$, define its normalized reciprocal by

$$h^\vee(x) = \operatorname{prim}\left(x^{\deg h} h(1/x)\right),$$

where the sign is chosen so that the leading coefficient is positive. We write $[h] = \{h, h^\vee\}$ for its reciprocal orbit. Since a package product is reciprocal, occurrence of either member implies occurrence of both in the product.

The following elementary estimate is independent of the $p$-adic theory developed in the later sections.

**Theorem 4.1** (Finite collision support of a fixed point)**.** *Fix $z \in \mathbb{C}^\times$ with $0 < r = |z| < 1$. If $L_m(z) = L_n(z)$, $0 < m < n$, and $k = n - m$, then*

$$\frac{k}{m} \leq \frac{2r^m}{1 - r^m}, \tag{4.1}$$

*and*

$$n \leq \frac{2m}{1 - r^m}. \tag{4.2}$$

*In particular, $\mathcal{G}_z$ has only finitely many edges, and those edges are effectively bounded in terms of $r$.*

*Proof.* The addition law and the collision $L_m = L_{m+k}$ give $L_m = z^m L_k$. Hence

$$k(1 - z^m) = m z^m (1 - z^k).$$

Taking absolute values yields

$$\frac{k}{m} = r^m \frac{|1 - z^k|}{|1 - z^m|} \leq r^m \frac{1 + r^k}{1 - r^m} \leq \frac{2r^m}{1 - r^m},$$

which is (4.1). Since $k \geq 1$, this forces $1 \leq 2mr^m/(1 - r^m)$; the right side tends to zero, so only finitely many $m$ occur. The original equality also gives

$$n|1 - z^m| = m|1 - z^n| \leq 2m,$$

and (4.2) follows from $|1 - z^m| \geq 1 - r^m$. □

For example, one may take

$$M(r) = \max\left\{m \geq 1 : \frac{2mr^m}{1 - r^m} \geq 1\right\},$$

with the convention $M(r) = 0$ when the set is empty. Then every edge has $m \leq M(r)$, followed by the explicit bound (4.2).

For a fixed package $\{c, d\}$, define

$$E_{c,d}(X) = \#\{\{a, b\} : a + b \leq X,\ \gcd(P_{a,b}, P_{c,d}) \neq 1\}.$$

**Theorem 4.2** (Finite support of factors; local finiteness)**.** *Every noncyclotomic reciprocal irreducible orbit $[h]$ occurs in only finitely many package products $P_{a,b}$. Consequently every package has only finitely many sharing partners. In particular, for each fixed $\{c, d\}$,*

$$E_{c,d}(X) = O_{c,d}(1).$$

*Proof.* Choose a member of $[h]$ having a root $\alpha$ with $0 < |\alpha| < 1$; this is possible because there are no unit-circle roots, and replacing $h$ by $h^\vee$ inverts all roots. If $[h]$ occurs in $P_{a,b}$, reciprocity of the product ensures that $\alpha$ is a root of $P_{a,b}$. It therefore gives one of the collision edges $\{a, a + b\}$ or $\{b, a + b\}$. Theorem 4.1 leaves only finitely many such edges and hence finitely many unordered packages.

A fixed package product has only finitely many irreducible reciprocal orbits. The union of their finite supports is exactly its set of sharing partners, so that set is finite. □

**Corollary 4.3** (Finite decidability of privacy). *For every fixed unordered pair $\{c, d\}$, whether it has a globally private factor is decidable by a finite exact calculation. More precisely, after factoring $P_{c,d}$, choose for each reciprocal factor orbit an algebraic root in the open unit disk and a certified rational upper bound $\rho < 1$ for its modulus. Replacing $r$ by $\rho$ in Theorem 4.1 gives an explicit finite list of packages, and exact polynomial gcds decide all remaining comparisons.*

**Remark 4.4.** The bounds obtained from a root modulus need not be practical or uniform in $c, d$. The point is logical: target (a), package by package, is no longer an intrinsically infinite comparison. A uniform proof that at least one orbit survives the finite comparison is still missing.

# 5 The divided-difference sign law

The next several sections derive analytic restrictions on any two-loop configuration. This one begins with a one-dimensional sign law: once a collision $L_m = L_n$ is fixed, the graph of $|1 - tA|^2 - |z|^{2t}$ crosses zero with a completely controlled sign pattern.

After replacing $z$ by $1/z$ and replacing every edge by its complementary edge, we may assume $0 < |z| < 1$. Put $R = |z|^2$.

**Theorem 5.1** (Sign law). *Suppose $L_m(z) = L_n(z) = A$, where $0 < m < n$ and $0 < |z| < 1$. Then for every real $t \notin \{0, m, n\}$,*

$$\operatorname{sgn}\bigl(|1 - tA|^2 - R^t\bigr) = \operatorname{sgn}\bigl(t(t - m)(t - n)\bigr).$$

*Proof.* Let $F_A(t) = |1 - tA|^2 - R^t$. We have $F_A(0) = F_A(m) = F_A(n) = 0$, and

$$F_A'''(t) = -(\log R)^3 R^t > 0.$$

The third divided-difference mean-value theorem gives

$$\frac{F_A(t)}{t(t - m)(t - n)} = F_A[t, 0, m, n] = \frac{F_A'''(\xi)}{3!} > 0$$

for some $\xi$ in the convex hull of the four nodes. □

In particular, the affine interpolation $1 - tA$ lies strictly inside the circle $|w| = |z|^t$ for $m < t < n$, and strictly outside it for $0 < t < m$ and $t > n$.

# 6 Exact tail conjugacy and the strict chord obstruction

Here the sign law becomes a geometric obstruction. A first collision conjugates the later tail of the sequence $L_k(z)$ to a rescaled copy centered at the gap; a second collision must therefore satisfy a strict chord condition. This is the main analytic mechanism used in the early reductions.

**Theorem 6.1** (Exact tail-conjugacy identity). *Assume*

$$L_m(z) = L_{m+a}(z) = A, \qquad m, a \geq 1.$$

*Then*

$$A = z^m L_a(z), \tag{6.1}$$

*and, for every integer $j \geq 1$,*

$$L_{m+j}(z) - A = \frac{j}{m+j} z^m (L_j(z) - L_a(z)). \tag{6.2}$$

*The identity extends to every positive real $j$ after choosing a logarithm of $z$ and defining $z^j$ continuously.*

*Proof.* Apply (3.1) with $(u, v) = (m, a)$:

$$(m + a)L_{m+a} = mL_m + az^m L_a.$$

Since $L_{m+a} = L_m$, this yields (6.1). Applying (3.1) with $(u, v) = (m, j)$, subtracting $(m + j)A$, and using (6.1), gives

$$(m + j)(L_{m+j} - A) = jz^m(L_j - L_a),$$

which is (6.2). □

Thus a loop on the interval $[m, m + a]$ conjugates the entire tail, after an affine translation and a simple scalar weight, to the original sequence centered at its gap $a$.

**Corollary 6.2** (Shifted one-crossing law)**.** *Under the hypotheses of Theorem 6.1, for every real $j > 0$,*

$$\operatorname{sgn}\left(|1 - jL_a(z)|^2 - |z|^{2j}\right) = \operatorname{sgn}(j - a). \tag{6.3}$$

*Proof.* From (6.1),

$$1 - (m + j)A = z^m(1 - jL_a).$$

Therefore

$$F_A(m + j) = |z|^{2m} \left(|1 - jL_a|^2 - |z|^{2j}\right).$$

For $j \neq a$, Theorem 5.1, with $n = m + a$, gives (6.3). At $j = a$, both sides are zero directly; here $\operatorname{sgn}(0) = 0$. □

Now suppose a second loop occurs later. Since the four endpoints are distinct, we may choose the smallest endpoint and write the configuration as

$$L_m = L_{m+a}, \qquad L_{m+s} = L_{m+t}, \qquad 0 < s < t. \tag{6.4}$$

**Corollary 6.3** (Strict chord constraint)**.** *Every configuration* (6.4) *satisfies*

$$\frac{s}{m+s}(L_s - L_a) = \frac{t}{m+t}(L_t - L_a), \tag{6.5}$$

*and therefore*

$$L_t = (1 - \lambda)L_a + \lambda L_s, \qquad \lambda = \frac{s(m + t)}{t(m + s)} \in (0, 1). \tag{6.6}$$

*Moreover, $L_s \neq L_a$, so $L_t$ lies in the open line segment joining $L_a$ and $L_s$.*

*Proof.* Apply (6.2) with $j = s, t$, then use $L_{m+s} = L_{m+t}$. This gives (6.5), and solving it gives (6.6). The inequalities $0 < \lambda < 1$ are equivalent to $0 < s < t$. The four edge endpoints are distinct, so $s \neq a$. If $L_s = L_a$, Lemma 3.3 produces $L_m = L_{m+s}$; together with $L_m = L_{m+a}$ this gives three distinct equal values, contradicting Theorem 3.1. □

There is an equivalent block-centroid form. Put $b = t - s$. The two collisions imply

$$z^s L_b = \frac{m}{m+s} L_a + \frac{s}{m+s} L_s. \tag{6.7}$$

Thus the average of the geometric block of length $b$ beginning at $s$ lies strictly between the two prefix averages $L_a$ and $L_s$, after the common factor $1 - z$ is removed.

Clearing denominators in (6.5) gives a useful sparse algebraic residue of the two-loop problem:

$$a(m+s)z^t - a(m+t)z^s - m(t-s)z^a + (a+m)(t-s) = 0. \tag{6.8}$$

Any surviving counterexample must satisfy both the original trinomial $H_{m,m+a}(z) = 0$ and the quadrinomial (6.8), as well as the strict segment condition (6.6) and the shifted sign law (6.3).

## 7 Phase localization, a positive cone, and the real-root subproblem

The chord condition still leaves complex phases to control. This section extracts explicit phase bounds from the trinomial equation, rewrites the quadrinomial obstruction as a positive-cone identity, and isolates the special order constraints that occur for negative real roots.

Write a collision as $L_m = L_{m+a}$, and let $z = re^{i\theta}$, $0 < r < 1$, with $\theta \in (-\pi, \pi]$. The trinomial equation has the exact form

$$z^m(m + a - mz^a) = a. \tag{7.1}$$

The denominator on the right of $z^m = a/(m + a - mz^a)$ lies in a disk centered at the positive real number $m + a$, of radius $mr^a$.

**Theorem 7.1** (Exact phase bounds)**.** *For every collision $L_m = L_{m+a}$ with $0 < |z| = r < 1$,*

$$\operatorname{dist}(m\theta, 2\pi\mathbb{Z}) \le \arcsin\left(\frac{mr^a}{m+a}\right). \tag{7.2}$$

*If a second collision is written as*

$$L_{m+s} = L_{m+t}, \qquad 0 < s < t,$$

*then, with $b = t - s$,*

$$\operatorname{dist}((m+s)\theta, 2\pi\mathbb{Z}) \le \arcsin\left(\frac{(m+s)r^b}{m+t}\right), \tag{7.3}$$

$$\operatorname{dist}(s\theta, 2\pi\mathbb{Z}) \le \arcsin\left(\frac{mr^a}{m+a}\right) + \arcsin\left(\frac{(m+s)r^b}{m+t}\right). \tag{7.4}$$

*If in addition $m < a$, the first bound is strictly smaller than $\pi/6$.*

*Proof.* The largest possible argument of a point in a disk centered at a positive real number $R$ with radius $\rho < R$ is $\arcsin(\rho/R)$. Apply this to (7.1). The second collision gives

$$z^{m+s}(m + t - (m+s)z^{t-s}) = t - s,$$

which proves (7.3). Circular distance to $2\pi\mathbb{Z}$ is subadditive, so subtracting the two multiples of $\theta$ gives (7.4). If $m < a$, then $mr^a/(m + a) < 1/2$. $\square$

**Remark 7.2** (Two useful normalizations cannot be conflated)**.** Replacing $z$ by $1/z$ swaps the lower endpoint and the gap of every edge. Thus one may orient a two-loop configuration so that its smallest entry among the two lowers and two gaps is a lower endpoint. This operation, however, also replaces $|z|$ by $|z|^{-1}$. Without an additional argument one cannot simultaneously impose that global-minimum normalization and $|z| < 1$. The stronger $\pi/6$ estimate is therefore conditional on $m < a$ in the inside orientation, not an automatic global normalization.

The quadrinomial in (6.8) has a useful decomposition. Continue to write $b = t - s$, and let its left side be $R_{m,a;s,t}(x)$.

**Proposition 7.3** (Positive-cone decomposition)**.** *One has the exact identity*

$$R_{m,a;s,t}(x) = \frac{a(m+t)}{t} H_{s,t}(x) + \frac{m(t-s)}{t} H_{a,t}(x). \tag{7.5}$$

*If $t > a$, then*

$$\frac{R_{m,a;s,t}(x)}{(x-1)^2}$$

*is a polynomial all of whose coefficients are strictly positive. At a surviving common root,*

$$\frac{H_{s,t}(z)}{H_{a,t}(z)} = -\frac{m(t-s)}{a(m+t)} \in \mathbb{R}_{<0}. \tag{7.6}$$

*Proof.* Expanding the right side of (7.5) gives (6.8). For $0 < u < v$,

$$\frac{H_{u,v}(x)}{(x-1)^2} = \sum_{j=0}^{u-1} (v-u)(j+1)x^j + \sum_{j=u}^{v-2} u(v-j-1)x^j, \tag{7.7}$$

so all coefficients are positive. When $t > a$, both kernels in (7.5) are of this form and both scalar coefficients are positive. At a root of $R_{m,a;s,t}$, neither kernel value can vanish: otherwise both would vanish and $L_a = L_s = L_t$, contrary to the no-three theorem (the indices are distinct in a genuine configuration). Dividing (7.5) therefore gives (7.6). $\square$

Thus the disjoint and interlacing interval patterns reduce to a concrete sector problem: conditional on $H_{m,m+a}(z) = 0$, the two kernel values in (7.6) would have to lie on opposite rays. The nested case $t < a$ remains algebraically different.

There is a further ordering theorem when the common root is real.

**Proposition 7.4** (Ordering of two hypothetical negative-real loops)**.** *Let $z = -r$, $0 < r < 1$. Every collision edge $\{m, n\}$ has $m$ even and $n$ odd. If two distinct collision edges satisfy $m < p$, then*

$$m < p, \qquad n < q, \qquad n - m > q - p. \tag{7.8}$$

*In particular, the later edge has larger total and smaller gap; the nested interval pattern is impossible.*

*Proof.* Put

$$E(X) = \frac{1 - e^{-X}}{X}, \qquad O(Y) = \frac{1 + e^{-Y}}{Y}.$$

Both functions are strictly decreasing on $(0, \infty)$. Values of $L_k(-r)$ on even and odd indices are respectively $uE(uk)$ and $uO(uk)$, where $u = -\log r$. Equal values cannot have the same parity;

an odd lower and an even upper are also impossible because $L_m(-r) > 1/m > 1/n > L_n(-r)$. Hence the lower index is even and the upper one odd.

For each $X > 0$, the equation $E(X) = O(Y)$ has a unique solution $Y > X$. Write it as $Y = \phi(X)$, and let the common value be $c$. Using

$$E'(X) = \frac{1 - c(X+1)}{X}, \qquad O'(Y) = \frac{1 - c(Y+1)}{Y},$$

implicit differentiation gives

$$0 < \phi'(X) = \frac{Y}{X}\frac{c(X+1) - 1}{c(Y+1) - 1} < 1;$$

the last inequality is equivalent to $(1-c)Y > (1-c)X$. Thus $\phi$ is increasing while $\phi(X) - X$ is strictly decreasing. Applying this at $X = um, up$ proves $n < q$ and $n - m > q - p$. □

This does not yet exclude two negative-real loops, but it reduces them to a strictly ordered tradeoff that is invisible in the general complex case.

# 8 The Schur deformation

The trinomial $H_{m,n}$ extends to a two-variable Schur deformation, whose $t = 1$ specialization recovers $H_{m,n}/(x-1)^2$. We record that deformation here; the next section expands it to second order in $t$ to compare the root branches of two colliding edges.

Define the bivariate deformation

$$\begin{aligned} \mathcal{F}_{m,n}(x,t) &= (1 - t^m)x^n + (t^n - 1)x^m + t^m - t^n \\ &= \det\begin{pmatrix} 1 & t^n & x^n \\ 1 & t^m & x^m \\ 1 & 1 & 1 \end{pmatrix}. \end{aligned} \tag{8.1}$$

It is divisible by $(t-1)(x-1)(x-t)$. Put

$$\mathcal{R}_{m,n}(x,t) = \frac{\mathcal{F}_{m,n}(x,t)}{(t-1)(x-1)(x-t)}.$$

The alternant formula for Schur polynomials gives

$$\mathcal{R}_{m,n}(x,t) = -s_{(n-2,m-1)}(1,t,x). \tag{8.2}$$

At $t = 1$,

$$\mathcal{R}_{m,n}(x,1) = -\frac{H_{m,n}(x)}{(x-1)^2}. \tag{8.3}$$

If $e = \gcd(m,n)$ and $[e](x) = (x^e - 1)/(x-1)$, the exact relation to our polynomial is

$$\mathcal{R}_{m,n}(x,1) = -Q_{m,n-m}(x)[e](x)^2. \tag{8.4}$$

Thus the specialization is literally $-Q$ in the primitive case and differs from it only by an explicit cyclotomic square in general.

# 9 The second jet of the Schur deformation

A tempting first-order transversality argument is false, because the relevant Schur curves have the same tangent at the specialization. The replacement is a second-order computation in symmetric coordinates; the explicit second jet is what survives for later use.

Concretely, the common tangent occurs at $t = 1$. The correct coordinates are

$$t = e^{\tau}, \qquad x = e^{\tau/2} y.$$

Writing $|\lambda|$ for the size of the partition, homogeneity and symmetry in the first two variables give

$$e^{-|\lambda|\tau/2} s_\lambda(1, e^{\tau}, e^{\tau/2} y) = s_\lambda(e^{-\tau/2}, e^{\tau/2}, y),$$

and the right side is even in $\tau$.

For our two-row partition define

$$S_{m,n}(y, \tau) = s_{(n-2,m-1)}(e^{-\tau/2}, e^{\tau/2}, y).$$

The bialternant formula yields

$$S_{m,n}(y, \tau) = \frac{\sinh(m\tau/2) y^n - \sinh(n\tau/2) y^m + \sinh((n-m)\tau/2)}{\sinh(\tau/2)(y^2 - 2\cosh(\tau/2) y + 1)}. \tag{9.1}$$

At $\tau = 0$, this is $H_{m,n}(y)/(y-1)^2$.

**Proposition 9.1** (Second-jet root branch)**.** *Suppose $z$ is a root of $H_{m,n}$ with $|z| \neq 1$, and put*

$$A = L_m(z) = L_n(z).$$

*Then the local root branch of $S_{m,n}(y, \tau) = 0$ through $(z, 0)$ has the form*

$$y_{m,n}(\tau) = z + \frac{z}{24}\left(m + n - \frac{3}{A}\right)\tau^2 + O(\tau^4). \tag{9.2}$$

*Proof.* Expanding (9.1) at $\tau = 0$ and using $H_{m,n}(z) = 0$ gives

$$S_{m,n}(z, \tau) = \frac{\tau^2}{24(z-1)^2}(m^3 z^n - n^3 z^m + (n-m)^3) + O(\tau^4). \tag{9.3}$$

Since $z^m = 1 - mA$ and $z^n = 1 - nA$, the bracket equals

$$mn(n-m)((m+n)A - 3). \tag{9.4}$$

On the other hand,

$$\begin{aligned} \frac{\partial}{\partial y} \frac{H_{m,n}(y)}{(y-1)^2}\bigg|_{y=z} &= \frac{mnz^{m-1}(z^{n-m} - 1)}{(z-1)^2} \\ &= -\frac{mn(n-m)A}{z(z-1)^2}. \end{aligned} \tag{9.5}$$

Here $A \neq 0$, because $A = 0$ would give $z^m = 1$, contrary to $|z| \neq 1$. Hence the derivative is nonzero. The implicit-function theorem gives a unique local root branch. Since $S_{m,n}(y, \tau) = S_{m,n}(y, -\tau)$, uniqueness forces $y_{m,n}(\tau) = y_{m,n}(-\tau)$; the branch is even. Combining this with (9.3)–(9.5) gives (9.2) with remainder $O(\tau^4)$. $\square$

If two edges $\{m,n\}$, $\{p,q\}$ share a root $z$, with collision values $A$, $B$, their comoving branches therefore separate as

$$y_{m,n}(\tau) - y_{p,q}(\tau) = \frac{z}{24}\left(m+n-p-q-3\left(\frac{1}{A}-\frac{1}{B}\right)\right)\tau^2 + O(\tau^4). \tag{9.6}$$

In the comoving local parameter $\tau$, the two root branches therefore differ by $O(\tau^2)$, and the order is exactly two unless the explicit bracket in (9.6) vanishes. For example, if $m+n=p+q$, higher contact would force $A=B$, contradicting the no-three theorem for disjoint edges. This second-order computation is the substitute for the failed first-order transversality argument.

## 10 A complete classification of rational roots

Rational roots are a boundary case that should not be left implicit. The positivity of the coefficient expansion makes the classification short, and the result is later used to rule out degree-one complementary factors in finite searches.

We first record the coefficient expansion, valid when $\gcd(a,b)=1$:

$$Q_{a,b}(x) = b\sum_{j=0}^{a-1}(j+1)x^j + a\sum_{j=a}^{a+b-2}(a+b-1-j)x^j. \tag{10.1}$$

In particular, the constant term is $b$, the leading coefficient is $a$, and all coefficients are positive.

**Theorem 10.1** (Rational-root classification)**.** *For arbitrary positive $a,b$, the polynomial $Q_{a,b}$ has a rational root if and only if*

$$(a,b)=(2,1) \quad \textit{or} \quad (a,b)=(1,2).$$

*The roots are respectively $-1/2$ and $-2$.*

*Proof.* First suppose $\gcd(a,b)=1$. By positivity of the coefficients there is no positive root. By reciprocity it suffices to classify a rational root $x=-r/s\in(-1,0)$, with $r,s$ coprime positive integers and $r<s$. Write $y=r/s\in(0,1)$ and $N=a+b$. If $a$ is odd and $N$ is even, then

$$H_{a,N}(-y) = ay^N + Ny^a + b > 0.$$

If $a$ and $N$ are both odd, then

$$H_{a,N}(-y) = -ay^N + Ny^a + b > 0,$$

because $y^a > y^N$ and $N>a$. If $a,N$ are both even, then

$$H_{a,N}(-y) = b - Ny^a + ay^N > 0;$$

its derivative is $aNy^{a-1}(y^{N-a}-1)<0$, and it decreases from $b$ to $0$. Hence only $a$ even and $N$ odd can remain; in particular $b$ is odd.

The rational-root theorem applied to (10.1) gives

$$r \mid b, \qquad s \mid a.$$

Writing $N = a + b$, the equation $H_{a,N}(-r/s) = 0$ becomes

$$bs^{a+b} = (a + b)r^a s^b + ar^{a+b}. \tag{10.2}$$

Let $\ell \mid s$, and put $t = v_\ell(s)$. Since $\ell \nmid b(a+b)r$, the two terms on the right of (10.2) have valuations $bt$ and $v_\ell(a)$, whereas the left side has valuation $(a + b)t > bt$. If the two right-hand valuations were unequal, the valuation of their sum would be their smaller value, strictly below that of the left side. They must therefore coincide, giving

$$v_\ell(a) = b\, v_\ell(s). \tag{10.3}$$

Likewise, if $\ell \mid r$ and $t = v_\ell(r)$, then the two right-hand valuations are $at$ and $(a + b)t$, while the left side has valuation $v_\ell(b)$. Hence

$$v_\ell(b) = a\, v_\ell(r). \tag{10.4}$$

Consequently,

$$s^b \mid a, \qquad r^a \mid b. \tag{10.5}$$

If $r \geq 2$, then $b \geq 2^a > a$, while $s \geq 2$ and (10.5) give $a \geq 2^b > b$, a contradiction. Hence $r = 1$.

Write $a = s^b A$ with $\gcd(A, s) = 1$. Dividing (10.2) by $s^b$ gives

$$b(s^a - 1) = A(s^b + 1) = a + A \leq \frac{3a}{2}. \tag{10.6}$$

For $a \geq 3$, the left side is at least $2^a - 1 > 3a/2$. Therefore $a = 2$, and (10.5) gives $s = 2$, $b = 1$, $A = 1$. Thus the only primitive root in the open unit disk is $-1/2$ for $Q_{2,1}$. Reciprocity gives $-2$ for $Q_{1,2}$.

Finally, for a nonprimitive pair write $a = gA$, $b = gB$. By (2.4), a rational root $x$ would make $x^g$ a rational root of a primitive $Q_{A,B}$. The only possibilities are $-1/2$ and $-2$. If $g > 1$, then $g$ must be odd, and the equality $g\, v_2(x) = \pm 1$ is impossible for rational $x$. Hence $g = 1$, and the two cases above are the only cases. □

This theorem closes the degree-one exception in the $p$-adic argument below.

## 11 Full additive-triple prime separation

We now move from analytic restrictions to arithmetic support. The goal is to show that primes appearing in the six entries of two additive triples cannot overlap except in precisely controlled ways.

We first isolate the two standard local-field facts used below; see, for example, [8, Chapters I–II].

**Lemma 11.1** (Two local-field criteria)**.** *Let $p$ be a prime.*

*(a) A root of unity of order prime to $p$ belongs to $\mathbb{Q}_p^{\mathrm{nr}}$. If $\zeta$ is such a root of unity and $v_p(\beta-\zeta) \notin \mathbb{Z}$, then $\mathbb{Q}_p(\beta)/\mathbb{Q}_p$ is ramified.*

*(b) A monic polynomial in $\mathbb{Z}_p[x]$ with unit constant term and square-free reduction has only unit roots, and every root lies in a finite unramified extension of $\mathbb{Q}_p$.*

*Proof.* For (a), prime-to-$p$ roots of unity are Teichmuller lifts and lie in the maximal unramified extension. If $\beta$ were also contained in an unramified extension, then the compositum containing $\beta$ and $\zeta$ would be unramified, whose normalized value group is $\mathbb{Z}$, contradicting the nonintegral valuation of their difference. Part (b) is the usual Hensel lifting description of finite etale $\mathbb{Z}_p$-algebras. Monicity makes every root integral, and the unit constant term forces all root valuations to be zero. $\square$

**Lemma 11.2** (Good package away from its additive triple)**.** *Let $p$ be odd. If $p \nmid cd(c+d)$, then every root of both orientations in $\mathcal{P}_{\{c,d\}}$ is a $p$-adic unit lying in a finite unramified extension of $\mathbb{Q}_p$.*

*Proof.* Write $c = hC, d = hD$, with $\gcd(C, D) = 1$. Then $p \nmid hCD(C + D)$. Borisov's discriminant formula [1, Lemma 2.1] shows that the primitive polynomial $Q_{C,D}$, after division by its unit leading coefficient, is monic over $\mathbb{Z}_p$, has unit constant term, and has square-free reduction. Furthermore,

$$\frac{d}{dx} Q_{C,D}(x^h) = hx^{h-1} Q'_{C,D}(x^h).$$

At a root modulo $p$, one has $x \neq 0$, $p \nmid h$, and the last factor is nonzero because the primitive reduction is square-free. Thus $Q_{C,D}(x^h)$ also has square-free reduction. Equation (2.4) differs from this composition only by a scalar. Lemma 11.1(b) applies, and the reciprocal orientation is identical. $\square$

For reference, Borisov's Lemmas 2.8–2.11 give the following consequence for the primitive polynomial $Q_{A,B} = f_{A+B,A,B}$.

**Lemma 11.3** (Ramification supplied by a primitive additive-triple prime)**.** *Let $\gcd(A, B) = 1$, let $N = A + B$, and suppose that $p \mid ABN$.*

*(a) If $p \geq 5$, every root of either orientation is either a nonunit or lies at nonintegral-valued distance from a root of unity of order prime to $p$. In the latter case its local field is ramified.*

*(b) If $p = 3$, each fixed orientation has at most one unit root in $\mathbb{Q}_3^{\mathrm{nr}}$; every other root is either a nonunit or ramified.*

*Proof.* The three numbers $A, B, N$ are pairwise coprime, so $p$ divides exactly one of them. If $p \mid N$, Lemmas 2.8–2.9 of [1] place all roots in clusters around prime-to-$p$ roots of unity. At nontrivial centers the successive distances have valuations

$$\frac{1}{p}, \ \frac{1}{p^2 - p}, \ldots,$$

and at the center 1 the first layer has valuation $1/(p - 2)$, followed by the same higher layers. For $p \geq 5$ all are nonintegral. For $p = 3$, the first layer at 1 consists of exactly one root at integral valuation; all other layers are fractional.

If $p \mid B$, Borisov's Lemma 2.10 gives $A$ nonunit roots and places the remaining roots in the same root-of-unity clusters. If $p \mid A$, Lemma 2.11 gives the reciprocal statement. Lemma 11.1(a) converts every fractional cluster distance into ramification. The same analysis applies after interchanging the two orientations. $\square$

**Theorem 11.4** (Full additive-triple prime separation). *Write $a = gA, b = gB$, with $\gcd(A, B) = 1$. Suppose*

$$p \mid AB(A + B), \qquad p \nmid cd(c + d). \tag{11.1}$$

*If $p \geq 5$, then*

$$\gcd_{\mathbb{Q}[x]}(P_{a,b}, P_{c,d}) = 1. \tag{11.2}$$

*If $g = 1$, the same conclusion holds for $p = 3$.*

*Proof.* Suppose an irreducible $h \in \mathbb{Q}[x]$ divides both package products. Fix an orientation of each package divisible by $h$, and choose a root $\alpha \in \overline{\mathbb{Q}}_p$ of $h$. Lemma 11.2 shows that $\alpha$ is a unit in an unramified extension. Put $\beta = \alpha^g$. Then $\beta$ is a root of the corresponding primitive orientation $Q_{A,B}$ or $Q_{B,A}$, and it is again a unit in an unramified extension. For $p \geq 5$, this contradicts Lemma 11.3(a).

Now let $p = 3$ and $g = 1$. Every root of $h$ is a root of the good second orientation and therefore lies in $\mathbb{Q}_3^{\mathrm{nr}}$. The fixed first orientation has at most one such unit root by Lemma 11.3(b). Since $h$ is separable, $\deg h \leq 1$. The rational-root classification (Theorem 10.1) forces the package $\{1, 2\}$. The same theorem would force the second package to be $\{1, 2\}$ as well, contrary to $3 \nmid cd(c+d)$. □

**Corollary 11.5** (Support compatibility). *Fix $\{c, d\}$ and let*

$$S = \{p : p \mid 6cd(c + d)\}.$$

*If $P_{a,b}$ shares a factor with $P_{c,d}$, and $a = gA, b = gB$ is its primitive reduction, then*

$$\mathrm{Supp}(AB(A + B)) \subseteq S. \tag{11.3}$$

*If both packages are primitive, their primitive additive-triple products have the same prime support outside $\{2, 3\}$.*

*Proof.* Every prime $p \geq 5$ in the left side but outside $S$ is excluded by Theorem 11.4; the primes $2, 3$ were inserted into $S$. For two primitive packages, apply the theorem in both directions. □

## 12 The $S$-unit reduction and finite primitive shapes

Once the relevant prime support is controlled, the possible primitive shapes become finite. This section packages that finiteness in the language of $S$-unit equations so that later counting arguments have a fixed arithmetic input.

The support statement is much stronger than smoothness of the primitive total: all three pairwise coprime entries $A, B, A + B$ are supported on one fixed finite set.

**Theorem 12.1** (Finite primitive-shape theorem). *Fix $\{c, d\}$ and let $S$ be as in Corollary 11.5. There are only finitely many ordered primitive pairs $(A, B)$ that can occur as primitive reductions of sharing partners. If $s = |S|$, their number is at most*

$$2^{16s+8}. \tag{12.1}$$

*Proof.* Let $N = A + B$, and put

$$u = \frac{A}{N}, \qquad v = \frac{B}{N}.$$

By (11.3), both $u$ and $v$ belong to the rational $S$-unit group $U_S$, and $u + v = 1$. The group $U_S \times U_S$ has rank $2s$. The theorem of Beukers–Schlickewei [3] bounds the number of solutions of $x + y = 1$ in a multiplicative subgroup of $(\mathbb{C}^\times)^2$ of rank $r$ by $2^{8r+8}$, giving (12.1). The map $(A, B) \mapsto (A/N, B/N)$ is injective on primitive ordered pairs, because each fraction is already in lowest terms. $\square$

There is an elementary final step from shapes to actual scaled packages.

**Proposition 12.2** (Finite scaling for a fixed shape)**.** *Fix a package $P_{c,d}$ and a primitive shape $(A, B)$. Only finitely many positive integers $g$ can make $P_{gA,gB}$ share a factor with $P_{c,d}$.*

*Proof.* A shared factor supplies a root $\alpha$ of $P_{c,d}$ such that $\beta = \alpha^g$ is a root of $P_{A,B}$. There are finitely many choices of $(\alpha, \beta)$, and $|\alpha| \neq 1$. For a fixed pair, $|\alpha|^g = |\beta|$ determines at most one positive integer $g$. $\square$

Theorem 12.1 and Proposition 12.2 give a second proof of local finiteness. Unlike the direct Archimedean proof, they also describe the arithmetic shape of the finite neighborhood. This is the main reason target (a) deserves renewed attention: failure of privacy for $P_{c,d}$ is now a finite divisibility assertion among explicitly constrained $S$-unit shapes and finitely many scalings.

# 13 Global proper-divisibility rigidity

A special danger is that one package might divide another after a composition. The results in this section rule out that kind of proper divisibility for primitive unequal pairs, preparing the self-power arguments that follow.

The next theorem also covers the boundary primitive shapes. Its principal algebraic input is an exact denominator calculation.

**Lemma 13.1** (Exact diagonal denominators)**.** *Let $a, b$ be coprime positive integers, put $N = a + b$, and set*

$$F(x) = 1 - \frac{N}{b}x^a + \frac{a}{b}x^N.$$

*For every $k \geq 0$,*

$$[x^{ka}]F(x)^{-1} = \frac{C_k}{b^k}, \tag{13.1}$$

*where*

$$C_k = N^k + \sum_{1 \leq r \leq \lfloor k/N \rfloor} \binom{k - br}{ar} N^{k-Nr}(-a)^{ar} b^{br}. \tag{13.2}$$

*In particular,* $\gcd(C_k, b) = 1$*, so the coefficient in* (13.1) *has denominator exactly* $b^k$*.*

*Proof.* Expand geometrically:

$$F(x)^{-1} = \sum_{j \geq 0} \left( \frac{N}{b}x^a - \frac{a}{b}x^N \right)^j.$$

If exactly $t$ of the $j$ selected monomials have exponent $N$, the resulting exponent is $(j - t)a + tN = ja + tb$. For this to equal $ka$, coprimality forces

$$t = ar, \qquad j = k - br$$

for an integer $r \geq 0$, and $t \leq j$ is equivalent to $Nr \leq k$. The corresponding contribution is

$$\binom{k-br}{ar}\frac{N^{k-Nr}(-a)^{ar}}{b^{k-br}}.$$

Putting all terms over $b^k$ gives (13.2). Every term with $r \geq 1$ is divisible by every prime dividing $b$, whereas $N^k \equiv a^k \not\equiv 0 \pmod p$ for $p \mid b$. Thus $C_k$ is coprime to $b$. $\square$

**Theorem 13.2** (Proper-divisibility rigidity)**.** *Let $a, b$ be coprime positive integers with $a \neq b$. For arbitrary positive $c, d$,*

$$Q_{a,b}(x) \mid Q_{c,d}(x) \quad \textit{in } \mathbb{Q}[x] \qquad \Longrightarrow \qquad (c,d) = (a,b). \tag{13.3}$$

*Proof.* If $c = d$, then $Q_{c,c} = c$ is constant and cannot be divisible by the nonconstant polynomial $Q_{a,b}$. If the unordered packages agree, the no-three theorem applied to the two complementary edges shows that the two orientations are coprime; hence the orientations in (13.3) must agree. We therefore assume that the packages are distinct, so Corollary 3.4 gives disjoint additive triples.

First suppose $a, b > 1$. The polynomial $Q_{a,b}$ is primitive in $\mathbb{Z}[x]$, because its leading and constant coefficients are the coprime integers $a, b$. Gauss's lemma gives

$$Q_{c,d} = Q_{a,b}R, \qquad R \in \mathbb{Z}[x].$$

Comparison of leading and constant coefficients gives

$$c = au, \qquad d = bv, \qquad u, v \in \mathbb{Z}_{>0}. \tag{13.4}$$

Disjointness of the additive triples gives $u, v \geq 2$.

Let $h = \gcd(c,d)$, let $[h](x) = (x^h - 1)/(x-1)$, and put $S(x) = [h](x)^2 R(x) \in \mathbb{Z}[x]$. Writing $N = a + b$, the identity between the trinomial numerators is

$$H_{c,c+d}(x) = H_{a,N}(x)S(x).$$

Reduction modulo the ideal $(x^c) = (x^{au})$ gives

$$bv \equiv b\left(1 - \frac{N}{b}x^a + \frac{a}{b}x^N\right)S(x) \pmod{x^{au}},$$

and hence

$$S(x) \equiv v\left(1 - \frac{N}{b}x^a + \frac{a}{b}x^N\right)^{-1} \pmod{x^{au}}. \tag{13.5}$$

For $0 \leq k < u$, the coefficient of $x^{ka}$ on the left is integral. Lemma 13.1 therefore gives $b^k \mid v$, and at $k = u - 1$,

$$b^{u-1} \mid v. \tag{13.6}$$

Reciprocating the divisibility and repeating the argument gives

$$a^{v-1} \mid u. \tag{13.7}$$

Thus $v \geq 2^{u-1}$ and $u \geq 2^{v-1}$. If, say, $u \leq v$, then $u \geq 2^{v-1} \geq 2^{u-1}$; for $u \geq 3$ this is impossible. Hence $u = 2$, and then $2 \geq 2^{v-1}$ gives $v = 2$. Thus $u = v = 2$. The assumed divisibility then gives $Q_{a,b}(x) \mid Q_{a,b}(x^2)$. Squaring maps the finite root set into itself, so every root is eventually periodic under squaring and hence a root of unity, contradicting Lemma 2.2. This eliminates the interior case.

It remains to treat a boundary shape. By reciprocity it is enough to assume $b = 1$ and $a > 1$. Comparison of leading coefficients gives

$$c = au \qquad (u \in \mathbb{Z}_{>0}). \tag{13.8}$$

Since the packages are distinct, additive-triple separation gives $d \geq 2$. Reciprocating yields

$$Q_{1,a} \mid Q_{d,c}.$$

Apply Lemma 13.1 to the source parameters $(1, a)$ and target parameters $(d, c)$. The target leading multiplier is $d$, and its constant multiplier is $u = c/a$; taking the diagonal coefficient with $k = d-1$ gives

$$a^{d-1} \mid u, \qquad \text{hence} \qquad c = au \geq a^d. \tag{13.9}$$

Now

$$Q_{a,1}(x) = 1 + 2x + \cdots + ax^{a-1}.$$

By the Eneström–Kakeya coefficient-ratio bound [4], every root $\alpha$ satisfies

$$r := |\alpha| \leq \frac{a-1}{a} \leq e^{-1/a} < 1. \tag{13.10}$$

The same $\alpha$ is a root of $Q_{c,d}$, so the collision estimate (4.1), applied to the edge $\{c, c+d\}$, gives

$$\frac{d}{c} \leq \frac{2r^c}{1-r^c}, \qquad d \leq \frac{2c}{e^{c/a}-1}.$$

The function $x/(e^{x/a}-1)$ is decreasing for $x > 0$: with $y = x/a$, the numerator of its derivative is $e^y(1-y) - 1 < 0$. Using (13.9) in (13), and then $e^y - 1 > y^2$ for $y \geq 2$, yields

$$d \leq \frac{2a^d}{e^{a^{d-1}}-1} < 2a^{2-d}.$$

For $d = 2$ the right side is strictly smaller than 2, and for $d \geq 3$ it is at most 1, a contradiction. Thus no distinct target exists. Reciprocity treats the case $a = 1$. $\square$

The orientation bookkeeping in the proof is as follows. If the unordered packages agree, the two reciprocal orientations are coprime by the no-three theorem, so a divisibility relation must preserve the selected orientation. If the packages are distinct, Corollary 3.4 gives disjoint scaled triples, and the coefficient comparison fixes the leading and constant roles separately. The boundary case is reduced by reciprocity before the diagonal-denominator argument is applied. Thus no step identifies $(a, b)$ with $(b, a)$ unless that reciprocal orientation has first been handled explicitly.

**Corollary 13.3** (Primitive irreducibility gives global isolation)**.** *Let $a, b$ be coprime and unequal. If $Q_{a,b}$ is irreducible over $\mathbb{Q}$, then the package $\{a, b\}$ is globally isolated from the entire primitive and nonprimitive family.*

*Proof.* The reciprocal $Q_{b,a}$ is irreducible as well. A common package factor would make one of these two polynomials divide one orientation of the other package, contradicting Theorem 13.2 unless the packages agree. $\square$

## 14 The Schinzel fibre bound, self-power rigidity, and isolation

This section proves the isolation theorem for irreducible primitive bases. The key point is that a shared factor in a lift would force an internal power relation for a root; the fibre bound and Kummer analysis exclude that relation.

The proper-divisibility theorem controls a whole primitive polynomial. We next make Schinzel's trinomial-fibre argument explicit, then combine it with a local Newton-edge obstruction and a Frobenius-quotient argument.

**Lemma 14.1** (Elementary binomial-fibre lemma)**.** *Let* $0 < m < n$*, let* $d \geq 1$*, and assume* $\gcd(\gcd(m,n), d) = 1$*. For every* $c \in \mathbb{C}^{\times}$*,*

$$\deg \gcd\big(H_{m,n}(x), x^d - c\big) \leq 1.$$

*Proof.* Suppose that the gcd has degree at least two. Since $x^d - c$ is separable, there are two distinct common roots $\alpha$ and $\zeta\alpha$, where $\zeta^d = 1$ and $\zeta \neq 1$. Put $g = \gcd(m,n)$. The hypothesis gives $\gcd(g,d) = 1$, so $\xi \mapsto \xi^g$ is an automorphism of $\mu_d$. Because

$$H_{m,n}(x) = gH_{m/g,n/g}(x^g),$$

the two transformed roots remain distinct, and it is enough to treat $\gcd(m,n) = 1$. Set $r = n - m$; then $m, r, n$ are pairwise coprime.

Write

$$u = \zeta^m, \qquad v = \zeta^n, \qquad X = \alpha^m, \qquad Y = \alpha^n.$$

The two root equations are

$$mY - nX = -r, \qquad mvY - nuX = -r.$$

If $u = v$, these equations give $u = v = 1$, and the coprimality of $m, n$ gives $\zeta = 1$, a contradiction. Hence

$$X = \frac{r(v-1)}{n(v-u)}, \qquad Y = \frac{r(u-1)}{m(v-u)}.$$

Since $X^n = Y^m$, one obtains the master identity

$$m^m r^r (\zeta^n - 1)^n = n^n \zeta^{mr} (\zeta^m - 1)^m (\zeta^r - 1)^r. \tag{14.1}$$

None of $\zeta^m, \zeta^n, \zeta^r$ is 1, by the displayed formulas and $\alpha \neq 0$.

Let $D = \operatorname{ord}(\zeta) = \ell^a s$, where $(\ell, s) = 1$. For each rational prime $\ell$, choose a place of $\mathbb{Q}(\zeta)$ above $\ell$ and normalize its valuation by $v_\ell(\ell) = 1$. The standard cyclotomic valuation formula is

$$v_\ell(1 - \zeta^k) = \begin{cases} 0, & s \nmid k, \\ \dfrac{1}{\varphi\big(\ell^{a - \min(a, v_\ell(k))}\big)}, & s \mid k,\ \zeta^k \neq 1. \end{cases}$$

It follows by separating the prime-to-$\ell$ Teichmuller part from the $\ell$-power part and using $\Phi_{\ell^j}(1) = \ell$.

Assume first that $\ell \mid m$, and put $t = v_\ell(m)$. Taking valuations in (14.1) gives

$$mt + nv_\ell(1 - \zeta^n) = mv_\ell(1 - \zeta^m) + rv_\ell(1 - \zeta^r). \tag{14.2}$$

If $s = 1$, then $a > t$, because $\zeta^m \neq 1$, and the valuation identity becomes

$$t + \frac{1}{\varphi(\ell^a)} = \frac{1}{\varphi(\ell^{a-t})},$$

or

$$t\varphi(\ell^a) + 1 = \ell^t.$$

This is impossible: since $a > t$, $t\ell^{a-1}(\ell - 1) + 1 > \ell^t$.

Suppose $s > 1$. Since $m, r, n$ are pairwise coprime, $s$ divides at most one of them. If it divides none, the valuation identity gives $mt = 0$. If $s \mid n$, its left side has an additional positive term and its right side has none. If $s \mid r$, then $mt = r/\varphi(\ell^a)$, so $m \mid r$, contrary to $(m, r) = 1$. Finally, if $s \mid m$, then

$$t = \frac{1}{\varphi(\ell^{a-t})},$$

so necessarily

$$\ell = 2, \qquad t = 1, \qquad a = 2.$$

If instead $\ell \mid n$, with $t = v_\ell(n)$, valuation of (14.1) gives

$$nv_\ell(1 - \zeta^n) = nt + mv_\ell(1 - \zeta^m) + rv_\ell(1 - \zeta^r).$$

When $s = 1$, division by $n = m + r$ gives exactly the same impossible identity

$$t + \frac{1}{\varphi(\ell^a)} = \frac{1}{\varphi(\ell^{a-t})}.$$

When $s > 1$, the prime-to-$\ell$ part $s$ divides at most one of $m, r, n$. If $s \nmid n$, the right side is strictly positive while the left side is zero; if $s \mid n$, the identity reduces to $t = 1/\varphi(\ell^{a-t})$. Thus again $\ell = 2$, $t = 1$, and $a = 2$.

Finally, if $\ell \mid r$, apply the already proved $\ell \mid m$ case to the reciprocal edge $(r, n)$: the identity

$$x^n H_{r,n}(x^{-1}) = H_{m,n}(x)$$

sends the two roots $\alpha, \zeta\alpha$ to $\alpha^{-1}, \zeta^{-1}\alpha^{-1}$, and $\zeta^{-1}$ has the same order as $\zeta$. Hence every prime divisor of $mrn$ would have to be 2, occurring to the first power. Pairwise coprimality and $n = m + r$ force $(m, r, n) = (1, 1, 2)$. But $H_{1,2}(x) = (x - 1)^2$, which cannot share two distinct roots with the separable polynomial $x^d - c$. This contradiction proves the lemma. $\square$

**Theorem 14.2** (Confluent Schinzel degree bound)**.** *Let $0 < m < n$, $0 < p < q$, and put*

$$d_1 = \gcd(m, n), \qquad d_2 = \gcd(p, q).$$

*If $\gcd(d_1, d_2) = 1$, then*

$$\deg \gcd\big(H_{m,n}, H_{p,q}\big) \leq \min\left\{\frac{n}{d_1}, \frac{q}{d_2}\right\}. \tag{14.3}$$

*Consequently, if a noncyclotomic irreducible polynomial $h$ of degree $e$ divides both trinomials, then*

$$e \leq \min\left\{\frac{n}{d_1}, \frac{q}{d_2}\right\} - 2. \tag{14.4}$$

*Proof.* Write $p = d_2p'$, $q = d_2q'$. As a polynomial in $y = x^{d_2}$, the second trinomial has degree $q'$, so over $\mathbb{C}$

$$H_{p,q}(x) = \lambda \prod_{j=1}^{q'} (x^{d_2} - c_j),$$

where the $c_j \neq 0$ are counted with multiplicity. Since $\gcd(d_1, d_2) = 1$, Lemma 14.1 applies to every fibre, and hence

$$\deg \gcd(H_{m,n}, H_{p,q}) \leq \sum_{j=1}^{q'} \deg \gcd(H_{m,n}, x^{d_2} - c_j) \leq q' = \frac{q}{d_2}.$$

Interchanging the two trinomials gives the other half of (14.3). Their only common roots of unity are at 1, with multiplicity two in each. Thus a common noncyclotomic factor of degree $e$, together with $(x-1)^2$, lies in the gcd, proving (14.4). Here the fibre estimate is the elementary Lemma 14.1, so the bound relies on no external result. $\square$

## 14.1 Two internal-power obstructions

The first lemma isolates the exact local mechanism used at primes dividing an additive triple.

**Lemma 14.3** (Newton-edge power obstruction)**.** *Let $L/\mathbb{Q}_p$ be unramified, let $\zeta \in L$ be a root of unity of order prime to $p$, and let $\beta$ be algebraic over $L$. Suppose that for some integer $e \geq 1$,*

$$v_p(\beta - \zeta) = \frac{1}{e}, \qquad [L(\beta) : L] \leq e.$$

*Then $\beta \notin L(\beta)^p$.*

*Proof.* The value $1/e$ in the value group forces the ramification index of $L(\beta)/L$ to be divisible by $e$. The degree hypothesis therefore forces equality throughout: the extension has degree and ramification index exactly $e$, and its value group is $e^{-1}\mathbb{Z}$.

Suppose $\beta = w^p$ with $w \in L(\beta)$. Frobenius is an automorphism of the residue field of $L$, so there is a Teichmuller root $\eta \in L$ with $\eta^p = \zeta$. Reduction modulo the maximal ideal gives $w \equiv \eta$. Write $w = \eta + \delta$; then $v_p(\delta) \geq 1/e$. In

$$\beta - \zeta = (\eta + \delta)^p - \eta^p = \sum_{j=1}^{p-1} \binom{p}{j} \eta^{p-j} \delta^j + \delta^p,$$

every intermediate term has valuation at least $1 + j/e > 1/e$, and the last has valuation at least $p/e > 1/e$. This contradicts $v_p(\beta - \zeta) = 1/e$. $\square$

The odd-prime arguments below need only the following root packet. It is best taken directly from Borisov's complete Newton-polygon description rather than inferred from a partial coefficient calculation.

**Lemma 14.4** (The center-1 packet in every odd-prime role)**.** *Let $A, B$ be coprime, put $N = A + B$, and let $p$ be odd. If $p^\kappa$ exactly divides one of $A, B, N$, then exactly $p - 2$ roots $\gamma$ of $Q_{A,B}$ satisfy*

$$v_p(\gamma - 1) = \frac{1}{p-2}.$$

*Consequently the Newton polygon of $Q_{A,B}(1 + Y)$ has a side of horizontal length $p - 2$ and slope $-1/(p-2)$.*

*Proof.* Under the dictionary

$$Q_{A,B} = f_{N,A,B}$$

for Borisov's notation $(a, b, c) = (N, A, B)$. If $p \mid N$, the asserted packet is exactly Lemma 2.9 of [1]. If $p \mid B$, Lemma 2.10 first separates the $A$ nonunit roots and states that the remaining roots occur in root-of-unity clusters exactly as in the total role; the cluster centered at 1 therefore contains the same $p-2$ roots. If $p \mid A$, Lemma 2.11 gives the reciprocal endpoint statement and the same center-1 packet. The final assertion follows from the standard Newton-polygon root–slope correspondence: roots with $v_p(\gamma - 1) = 1/(p-2)$ contribute a side of slope $-1/(p-2)$, and their number is its horizontal length. $\square$

For an integer $t$ and a prime $p$, put

$$q_p(t) = \frac{t^p - t}{p}, \qquad C_p(X, Y) = \frac{(X+Y)^p - X^p - Y^p}{p}.$$

Both are integral. The following factor-level theorem is stronger than what is needed for an irreducible whole polynomial.

**Theorem 14.5** (Good-prime Frobenius bound for a proper factor)**.** *Let $A, B$ be coprime positive integers, put $N = A + B$, and let $h$ be an irreducible factor of $F = Q_{A,B}$ of degree $e$. Let $\beta$ be a root of $h$. If $p$ is an odd prime with*

$$p \nmid ABN$$

*and $\beta \in \mathbb{Q}(\beta)^p$, then*

$$e \le p - 2. \tag{14.5}$$

*Proof.* Choose $h \in \mathbb{Z}[x]$ primitive with positive leading coefficient and write $F = hk$ in $\mathbb{Z}[x]$. The leading and constant coefficients of $h$ are coprime, because they divide the coprime integers $A$ and $B$. Suppose $\beta = w^p$ in $K = \mathbb{Q}(\beta)$. Then $K = \mathbb{Q}(w)$, and the norm identity gives

$$\mathrm{lc}(h) = u^p, \qquad h(0) = v_0^p$$

for coprime integers $u > 0$ and $v_0$, where the sign is absorbed into $v_0$ because $p$ is odd.

Let $G \in \mathbb{Z}[x]$ be the primitive minimal polynomial of $w$, with positive leading coefficient $t$, and put $s = G(0)$. Since $G \mid h(x^p)$, one has $t \mid u^p$ and $s \mid v_0^p$, hence $(t, s) = 1$. The norm identity gives

$$\left(\frac{s}{t}\right)^p = \left(\frac{v_0}{u}\right)^p.$$

The odd $p$-th power map is injective on $\mathbb{Q}$, and both fractions are reduced. Therefore

$$t = u, \qquad s = v_0.$$

In particular, $p \nmid u$, because $u^p \mid A$.

Let $w_1, \ldots, w_e$ be the conjugates of $w$. Since $\mathbb{Q}(w) = \mathbb{Q}(\beta)$, the multiset $\{w_i^p\}$ is exactly the multiset of conjugates of $\beta$. Since $G/u$ is monic over $\mathbb{Z}_{(p)}$, the $w_i$ are integral over $\mathbb{Z}_{(p)}$. In the integral closure $\mathcal{O}$ of $\mathbb{Z}_{(p)}$,

$$\begin{aligned} G(x)^p &= u^p \prod_i (x - w_i)^p \\ &\equiv u^p \prod_i (x^p - w_i^p) = h(x^p) \pmod{p\mathcal{O}[x]}. \end{aligned}$$

Both sides have rational coefficients, and $p\mathcal{O} \cap \mathbb{Q} = p\mathbb{Z}_{(p)}$, so this is a coefficientwise congruence in $\mathbb{Z}_{(p)}[x]$. Coefficientwise freshman's dream also gives $G(x)^p \equiv G(x^p)$ and $h(x)^p \equiv h(x^p) \pmod p$. Hence $\overline{G(x^p)} = \overline{h(x^p)}$; injectivity of $f(x) \mapsto f(x^p)$ on $\mathbb{F}_p[x]$ yields

$$\overline{G} = \overline{h}.$$

Define

$$\Phi_p(f)(x) = \frac{f(x)^p - f(x^p)}{p}.$$

Because $u$ is a $p$-adic unit, the quotient

$$R = \mathbb{Z}_{(p)}[x]/(G) = \mathbb{Z}_{(p)}[x]/(G/u)$$

is a free, hence torsion-free, $\mathbb{Z}_{(p)}$-module of rank $e$. Since $\overline{G} = \overline{h}$, write $G = h + pE$ with $E \in \mathbb{Z}_{(p)}[x]$. In $R$, one has

$$h = -pE, \qquad h(x^p) = 0,$$

the second equality following from $G \mid h(x^p)$. Therefore

$$\Phi_p(h) = (-1)^p p^{p-1} E^p \in pR.$$

Since $R/pR \simeq \mathbb{F}_p[x]/(\overline{h})$, this proves

$$\overline{h} \mid \overline{\Phi_p(h)}.$$

The product rule

$$\Phi_p(fg) = f^p \Phi_p(g) + g(x^p)\Phi_p(f)$$

transfers this vanishing from $h$ to $F$. Applying the same rule to

$$H(x) = (x-1)^2 F(x) = Ax^N - Nx^A + B$$

transfers it from $F$ to $H$: at a root $\alpha$ of $\overline{h}$, the second product-rule term contains $F(\alpha^p) = F(\alpha)^p = 0$ in characteristic $p$. Thus

$$\overline{\Phi_p(H)}(\alpha) = 0 \quad \text{for every root } \alpha \text{ of } \overline{h}.$$

Set

$$U = A\alpha^B, \qquad V = B\alpha^{-A}.$$

The equation $H(\alpha) = 0$ gives $U + V = N$. Put

$$q_p(t) = \frac{t^p - t}{p}, \qquad C_p(X,Y) = \frac{(X+Y)^p - X^p - Y^p}{p}.$$

To make the coefficient calculation explicit, write

$$H = Ax^N - Nx^A + B.$$

In $\Phi_p(H)$, the mixed terms from the three summands contribute $x^{pA} C_p(U,V)$, while the coefficient corrections contribute

$$x^{pA}\left(\frac{q_p(A)}{A}U^p + \frac{q_p(B)}{B}V^p - q_p(N)\right).$$

Using $V = N - U$ gives

$$\alpha^{-pA}\overline{\Phi_p(H)}(\alpha) = T_{p,A,B}(U), \tag{14.6}$$

where

$$T_{p,A,B}(U) = C_p(U, N-U) + \frac{q_p(A)}{A}U^p + \frac{q_p(B)}{B}(N-U)^p - q_p(N) \in \mathbb{F}_p[U]. \tag{14.7}$$

The hypothesis $p \nmid ABN$ is essential here. The polynomial has degree at most $p$, and the identity

$$q_p(N) = q_p(A) + q_p(B) + C_p(A, B)$$

gives

$$T_{p,A,B}(A) = T'_{p,A,B}(A) = 0, \qquad T''_{p,A,B}(A) = A^{-1} + B^{-1} = \frac{N}{AB} \neq 0.$$

Thus $T_{p,A,B}$ is nonzero and $U = A$ is an exactly double root.

It remains to count the other roots. The reduction of $F$ is square-free. Indeed, a common root of $F$ and $F'$ would be a common root of $H$ and

$$H'(x) = ANx^{A-1}(x^B - 1).$$

It would satisfy $x^A = x^B = 1$, hence $x = 1$, whereas

$$F(1) = \frac{ABN}{2} \not\equiv 0 \pmod{p}.$$

Therefore $\overline{h}$ has $e$ distinct roots. The map

$$\alpha \longmapsto U = A\alpha^B$$

is injective on roots of $F$: equality of $U$ also gives equality of $V = N - U$, so the quotient of two roots has both its $A$-th and $B$-th powers equal to 1. Moreover $U = A$ would force $\alpha^A = \alpha^B = 1$, hence $\alpha = 1$, which is not a root. Consequently the $e$ roots of $\overline{h}$ give $e$ distinct roots of $T_{p,A,B}$ outside the exactly double root $A$. The remaining root-multiplicity budget is at most $p-2$, proving (14.5). □

**Theorem 14.6** (Self-power rigidity)**.** *Let $A, B$ be coprime positive integers with $A \neq B$, and suppose $F = Q_{A,B}$ is irreducible. If $\beta$ is a root and $K = \mathbb{Q}(\beta)$, then*

$$\beta \notin K^p \qquad \textit{for every prime } p. \tag{14.8}$$

*Proof.* Suppose $\beta = w^p$ with $w \in K$. Put $N = A + B$ and $D = N - 2$. Since the monic minimal polynomial of $\beta$ is $F/A$,

$$\mathrm{N}_{K/\mathbb{Q}}(w)^p = \mathrm{N}_{K/\mathbb{Q}}(\beta) = (-1)^D \frac{B}{A}. \tag{14.9}$$

For odd $p$, coprimality implies

$$A = u^p, \qquad B = v^p$$

for coprime positive integers $u, v$. For $p = 2$, equation (14.9) forces $D$ even and $A, B$ to be coprime squares; hence both are odd and $v_2(N) = 1$.

First suppose $p \mid ABN$. For odd $p$, Lemma 14.4 provides in each of the three possible roles a cluster at 1 containing exactly $e = p - 2$ roots $\gamma$ with

$$v_p(\gamma - 1) = \frac{1}{e}.$$

For $p = 2$, the preceding parity conclusion gives $v_2(N) = 1$. Since $A, B$ are distinct coprime odd squares, $N \geq 10$, so $N/2 > 1$; Borisov's Lemma 2.8 provides a nontrivial prime-to-2 center $\zeta$ with exactly $e = 2$ roots satisfying $v_2(\gamma - \zeta) = 1/2$.

Because $F$ is irreducible, a $\mathbb{Q}$-embedding of $K$ into $\overline{\mathbb{Q}}_p$ may send $\beta$ to one of these roots. Let $L$ be the unramified field generated by the cluster center. Every $L$-embedding fixes the cluster center and preserves the unique extension of the $p$-adic valuation, so $v_p(\sigma(\beta) - \zeta) = v_p(\beta - \zeta)$. Thus every $L$-conjugate remains on the same edge, which contains only $e$ roots; hence $[L(\beta) : L] \leq e$. Lemma 14.3 contradicts $\beta = w^p$.

It remains that $p \nmid ABN$. Then $p$ is odd, and Theorem 14.5, applied to the whole irreducible polynomial of degree $D$, gives $D \leq p - 2$. But $A = u^p$, $B = v^p$, and $u \neq v$, so

$$D = A + B - 2 \geq 1 + 2^p - 2 = 2^p - 1 > p - 2,$$

a contradiction. This proves (14.8). □

## 14.2 Kummer reduction and its complete elimination

**Lemma 14.7** (Degree under extraction of a root)**.** *Let $f \in \mathbb{Q}[x]$ be irreducible of degree $D$, let $r \geq 1$, and let $w$ be algebraic with $f(w^r) = 0$. Put $\beta = w^r$. Then*

$$[\mathbb{Q}(w) : \mathbb{Q}] = [\mathbb{Q}(w) : \mathbb{Q}(\beta)]D.$$

*In particular, equality of the two degrees is equivalent to $\mathbb{Q}(w) = \mathbb{Q}(\beta)$.*

*Proof.* This is the tower law for $\mathbb{Q}(\beta) \subseteq \mathbb{Q}(w)$, together with the irreducibility of $f$. □

**Theorem 14.8** (Exact Kummer reduction)**.** *Let $A, B$ be coprime and unequal, put $N = A + B$, and suppose $Q_{A,B}$ is irreducible. Assume that $\{rA, rB\}$ shares a factor with $\{sC, sD\}$, where $\gcd(C, D) = 1$. Independently interchange $A, B$ and $C, D$, as necessary, so that the selected common root lies in $Q_{rA,rB}$ and $Q_{sC,sD}$. Put*

$$g = \gcd(r, s), \qquad r_0 = r/g, \qquad s_0 = s/g.$$

*Then either the two unordered packages agree, or $r_0 > 1$ and a root $\beta$ of $Q_{A,B}$ is an $r_0$-th power inside $\mathbb{Q}(\beta)$.*

*Proof.* Let $\alpha$ be the common root and set $w = \alpha^g$. Then

$$Q_{A,B}(w^{r_0}) = 0, \qquad Q_{C,D}(w^{s_0}) = 0.$$

Let $k_w$ be the minimal polynomial of $w$, and put $M = C + D$. Because $|w| = |\alpha|^g \neq 1$, the polynomial $k_w$ is noncyclotomic. It divides both normalized package polynomials

$$Q_{A,B}(x^{r_0}), \qquad Q_{C,D}(x^{s_0}).$$

The associated exponent-pair gcds are $r_0, s_0$, which are coprime. Theorem 14.2 gives

$$\deg k_w \leq \min(N, M) - 2 \leq N - 2.$$

On the other hand, $\beta = w^{r_0}$ is a root of the irreducible polynomial $Q_{A,B}$ of degree $N - 2$, so Lemma 14.7 gives the reverse inequality. Hence $\mathbb{Q}(w) = \mathbb{Q}(\beta)$.

If $r_0 = 1$, then $Q_{A,B}$ divides $Q_{s_0C,s_0D}$, and Theorem 13.2 forces equality of the oriented pairs and hence of the packages. If $r_0 > 1$, the field equality gives $w \in \mathbb{Q}(\beta)$ and $w^{r_0} = \beta$, as asserted. □

**Theorem 14.9** (Global isolation of every lift of an irreducible shape)**.** *Let $A, B$ be coprime and unequal, and suppose $Q_{A,B}$ is irreducible. Then for every $r \geq 1$, the package $\{rA, rB\}$ is globally isolated.*

*Proof.* An unequal-package collision would give the second alternative in Theorem 14.8. Choose a prime $p \mid r_0$; then $\beta = (w^{r_0/p})^p$ inside $\mathbb{Q}(\beta)$, contradicting Theorem 14.6. $\square$

**Corollary 14.10** (Individual composition irreducibility also isolates)**.** *Let $A, B$ be coprime and unequal, and $r \geq 1$. If $Q_{A,B}(x^r)$ is irreducible over $\mathbb{Q}$, then $\{rA, rB\}$ is globally isolated.*

*Proof.* If a common factor existed, a source root $\alpha$ would have degree $r(A + B - 2)$. With the notation in the preceding proof, $[\mathbb{Q}(\alpha) : \mathbb{Q}(w)] \leq g$, so

$$[\mathbb{Q}(w) : \mathbb{Q}] \geq r_0(A + B - 2).$$

The Schinzel bound is at most $A + B - 2$, forcing $r_0 = 1$; proper-divisibility rigidity then forces equality of the packages. $\square$

**Theorem 14.11** (Density-one global isolation)**.** *Let $I(X)$ denote the number of unordered pairs $1 \leq a < b$, $a + b \leq X$, whose reciprocal packages are not globally isolated. Then*

$$I(X) \ll X^{20/11} \log(2X). \tag{14.10}$$

*Proof.* Write uniquely

$$a = rA, \qquad b = rB, \qquad \gcd(A, B) = 1, \qquad A < B.$$

By Theorem 14.9, a non-isolated package must have reducible primitive base $Q_{A,B}$.

To connect precisely with Borisov's notation, his triple is

$$a_{\mathrm{Bor}} = A + B, \qquad b_{\mathrm{Bor}} = A, \qquad c_{\mathrm{Bor}} = B.$$

For a fixed small $\varepsilon > 0$, the contrapositive of his Theorem 3.8 places every sufficiently large reducible primitive shape in the complementary numerical range counted by Theorem 3.9 [1]. The finitely many small totals are absorbed into the constant. Thus the number $R(Y)$ of reducible primitive shapes of total at most $Y$ satisfies

$$R(Y) \ll Y^{20/11} \log(2Y). \tag{14.11}$$

Therefore

$$I(X) \leq \sum_{r \leq X/3} R(X/r) \ll X^{20/11} \log(2X) \sum_{r \geq 1} r^{-20/11} \ll X^{20/11} \log(2X).$$

Since the total number of unordered unequal pairs of total at most $X$ is asymptotic to $X^2/4$, a density-one set of package vertices is globally isolated. $\square$

**Remark 14.12.** The estimate is global, not a bounded-box assertion: every certified package is coprime to every distinct package in the entire infinite family. The remaining set is merely the set not eliminated by the present proof; the theorem does not assert that its members actually share factors.

## 15 Factor-level power sieves, residual splitting, and exact signatures

Section 14 handles irreducible primitive bases. For reducible bases, the same power questions must be tracked factor by factor. The purpose here is to record exactly where endpoint primes can still create internal powers and which congruence signatures survive.

The local cluster argument constrains proper factors without assuming irreducibility of the whole primitive polynomial. We first retain the radical sieve, then resolve the equal-slope 2-adic block by its residual polynomial.

**Theorem 15.1** (Primitive total-radical sieve)**.** *Let $A, B$ be coprime positive integers, put $N = A+B$, and let $h \in \mathbb{Q}[x]$ be an irreducible factor of $Q_{A,B}$ of degree $e$. For every prime $p \mid N$,*

$$e \equiv 0 \quad \text{or} \quad -2 \pmod{p}. \tag{15.1}$$

*Consequently,*

$$\operatorname{rad}(N) \mid e(e+2). \tag{15.2}$$

*If the same reciprocal irreducible orbit occurs in primitive packages of totals $N$ and $M$, then*

$$\operatorname{lcm}(\operatorname{rad} N, \operatorname{rad} M) \mid e(e+2). \tag{15.3}$$

*Proof.* Fix $p^k \| N$. Since $p \nmid AB$, normalize a rational factor to be monic over $\mathbb{Z}_p$ with unit constant term, and work over the unramified extension containing the prime-to-$p$ cluster centers.

Assume first that $p$ is odd. At every nontrivial center, Borisov's Lemma 2.8 gives nonhorizontal Newton edges of lengths

$$p, \quad p^2 - p, \quad \dots, \quad p^k - p^{k-1},$$

with reduced slope denominators equal to those lengths. At the center 1, Lemma 14.4 gives a side of length $p-2$ and reduced slope denominator $p-2$; Borisov's Lemma 2.9 gives the subsequent $p$-divisible lengths. A monic local factor has a Newton polygon with integral vertices. Its horizontal contribution on an edge of reduced slope denominator $\ell$ is divisible by $\ell$; here $\ell$ equals the full edge length, so the factor takes that edge wholly or not at all. Contributions at nontrivial centers are divisible by $p$, while the center 1 contributes either 0 or $p-2$ modulo $p$. This proves (15.1) for odd $p$.

For $p = 2$, one must not separate Borisov's first two formal layers when $k \geq 2$: both have slope $1/2$ and form a single canonical side of length four. Its reduced slope denominator is nevertheless 2, so any local factor takes an even number of roots from that side. Every later nontrivial side has even reduced slope denominator, and the same is true for every nonhorizontal side at the center 1. Hence every factor degree is even. This is precisely (15.1) modulo 2.

Thus every $p \mid N$ divides $e(e+2)$, proving (15.2). For a common reciprocal orbit, apply the result to whichever of $h, h^\vee$ divides the second orientation; the reciprocal has the same degree. This gives (15.3). □

**Corollary 15.2** (A numerical coprimality criterion)**.** *Let two primitive packages have totals $N, M$, put*

$$\nu = \min(N, M), \qquad \rho = \operatorname{lcm}(\operatorname{rad} N, \operatorname{rad} M).$$

*If*

$$\rho > \nu(\nu - 2), \tag{15.4}$$

*then the packages are coprime. In particular, distinct primitive packages with coprime squarefree totals are coprime.*

*Proof.* A common factor has degree $1 \le e \le \nu - 2$, so $e(e+2) \le \nu(\nu-2)$, contradicting (15.3)–(15.4). For coprime squarefree totals, $\rho = NM > \nu(\nu - 2)$. $\square$

### 15.1 The merged 2-adic side is residual-separable

We use the first-order theorem of the residual polynomial in the following standard form; see [9, Theorem 1.21].

**Lemma 15.3** (Newton–Ore residual splitting)**.** *Let $L/\mathbb{Q}_2$ be unramified and let a polynomial over $L$ have a Newton side of slope $-1/2$, horizontal length four, and separable quadratic residual polynomial. After a finite unramified extension $L'/L$, the four roots belonging to that side split into two packets of two. Each root $\gamma$ in either packet satisfies*

$$[L'(\gamma) : L'] \le 2.$$

*Proof.* The denominator of the reduced slope is 2, and the side has residual degree two. After an unramified extension splitting the separable residual polynomial, its two distinct linear residual factors give, by the theorem of the residual polynomial, two factors of degree 2. Each associated root therefore has local degree at most two over the enlarged unramified base. $\square$

**Proposition 15.4** (Residual polynomials of the merged block)**.** *Let $A, B$ be coprime, $N = A + B$, and suppose a nontrivial prime-to-2 cluster center exists.*

*(a) If $N = 2^\kappa n$, $\kappa \ge 2$, $n$ odd, and $\zeta \ne 1$ is an $n$-th root of unity, then the length-four slope-$1/2$ side at $\zeta$ has residual polynomial*

$$R_N(Z) = (1 - \bar{\zeta}^A) + \bar{\zeta}^{-2} Z + \bar{\zeta}^{-4} Z^2.$$

*(b) If $B = 2^\kappa b$, $\kappa \ge 2$, $b$ odd, and $\zeta \ne 1$ is a $b$-th root of unity, then the corresponding unit side has residual polynomial*

$$R_B(Z) = (1 - \bar{\zeta}^A) + \bar{\zeta}^{A-2} Z + \bar{\zeta}^{A-4} Z^2.$$

*Both polynomials are separable. The case $2^\kappa \| A$ follows by reciprocity.*

*Proof.* Work in the unramified field generated by $\zeta$, and write

$$H_{A,N}(\zeta + Y) = \sum_{j \ge 0} c_j Y^j.$$

At a nontrivial center the denominator $(x-1)^2$ is a unit, so the relevant Newton data are unchanged on passing to $Q_{A,B}$.

In case (a), $\zeta^N = 1$, $A$ is odd, and

$$c_0 = N(1 - \zeta^A),$$

$$c_2 = \frac{AN}{2} \zeta^{-2} ((N-1) - (A-1)\zeta^A),$$

$$c_4 = A \binom{N}{4} \zeta^{-4} - N \binom{A}{4} \zeta^{A-4}.$$

The three valuations are respectively $\kappa, \kappa-1, \kappa-2$. After division by the powers of two on the side and reduction, the coefficients are

$$1-\bar{\zeta}^A, \qquad \bar{\zeta}^{-2}, \qquad \bar{\zeta}^{-4}.$$

Here the second term in $c_4$ has larger valuation, while $2^{2-\kappa}\binom{N}{4}$ is odd.

In case (b), $\zeta^B = 1$, and direct cancellation gives

$$c_0 = B(1-\zeta^A), \qquad c_2 = \frac{ANB}{2}\zeta^{A-2},$$

$$c_4 = \frac{ANB}{24}(N^2 + NA + A^2 - 6(N+A) + 11)\zeta^{A-4}.$$

Since $A, N$ are odd and $N \equiv A \pmod 4$, the integer in parentheses is congruent to $2 \pmod 4$. The normalized residual coefficients are therefore

$$1-\bar{\zeta}^A, \qquad \bar{\zeta}^{A-2}, \qquad \bar{\zeta}^{A-4}.$$

Borisov's polygon identifies these points as the merged canonical side in both cases. In characteristic two,

$$R_N'(Z) = \bar{\zeta}^{-2} \neq 0, \qquad R_B'(Z) = \bar{\zeta}^{A-2} \neq 0.$$

Thus both residual polynomials are separable. □

**Corollary 15.5** (Internal powers at every total prime)**.** *Let $h$ be an irreducible factor of a primitive $Q_{A,B}$, and let $\beta$ be a root. If $p \mid A+B$, then*

$$\beta \notin \mathbb{Q}(\beta)^p.$$

*This includes $p = 2$ for every value of $v_2(A+B)$.*

*Proof.* For odd $p$, a root lies on a Newton side of some length $\ell$ and slope denominator $\ell$. Borisov's exact count gives at most $\ell$ roots at that center and distance, while the valuation denominator forces the local ramification degree to be at least $\ell$. Lemma 14.3 applies.

For $p = 2$, sides of length two with no equal-slope merger are handled in the same way. If the primitive total has odd part greater than one and $v_2(A+B) \geq 2$, Proposition 15.4(a) and Lemma 15.3 split the four-root side, after an unramified extension, into two degree-two packets. Lemma 14.3 then applies to each packet. If the primitive total is a pure power of two, there are no nontrivial centers; at the center 1, the nonzero sides have successive lengths $2, 4, 8, \ldots$ and distinct slopes, so the original edge argument applies directly. These cases exhaust all roots. □

## 15.2 Endpoint and good-prime power sieves

**Theorem 15.6** (Endpoint-prime internal-power sieve)**.** *Let $h$ be an irreducible factor of primitive $Q_{A,B}$, of degree $e$, and let $\beta$ be a root.*

*(a) Suppose $p^\kappa \| B$, for an arbitrary prime $p$. If $\beta \in \mathbb{Q}(\beta)^p$, then*

$$e \leq A, \qquad pA \mid e\kappa. \tag{15.5}$$

*(b) Suppose $p^\kappa \| A$. Then an internal $p$-th power forces*

$$e \leq B, \qquad pB \mid e\kappa. \tag{15.6}$$

*Equivalently, in (a),*

$$\frac{pA}{\gcd(pA,\kappa)} \mid e;$$

*in particular $\kappa < p$ is impossible. The reciprocal statement holds in (b).*

*Proof.* We prove (a); reciprocity gives (b). Borisov's endpoint description [1, Lemma 2.10] gives exactly $A$ nonunit roots, each of valuation $\kappa/A$. Every other root lies in a prime-to-$p$ root-of-unity cluster; the reciprocal statement uses [1, Lemma 2.11].

For odd $p$, every unit-cluster side has exact root count equal to its reduced slope denominator, so Lemma 14.3 excludes an internal $p$-th power there. For $p = 2$, the unmerged sides are treated identically. If $\kappa \geq 2$ and the odd part of $B$ is greater than one, Proposition 15.4(b) and Lemma 15.3 split the merged side into degree-two packets. If the odd part is one, there are no nontrivial centers and the center-1 sides have distinct slopes. Thus unit-cluster roots are excluded for every $p$ and every $\kappa$.

If $\beta = w^p$, every rational conjugate of $\beta$ is likewise a $p$-th power in its conjugate field. Hence all $e$ roots of $h$ must belong to the nonunit block, proving $e \leq A$. The leading coefficient of $h$ is a $p$-adic unit, and therefore

$$v_p\Big(N_{\mathbb{Q}(\beta)/\mathbb{Q}}(\beta)\Big) = \frac{e\kappa}{A}.$$

The norm is a rational $p$-th power, so this integer is divisible by $p$. Thus $pA \mid e\kappa$. If $\kappa < p$, then $e\kappa \leq A\kappa < pA$, a contradiction. ☐

**Corollary 15.7** (Good-prime internal-power sieve)**.** *Under the hypotheses of Theorem 14.5,*

$$\operatorname{rad}(A+B) \leq e(e+2) \leq p(p-2).$$

*In particular,* $\operatorname{rad}(A+B) > p(p-2)$ *rules out an internal $p$-th power for every factor.*

*Proof.* Combine Theorems 15.1 and 14.5. ☐

We next refine the Kummer trigger prime by prime. We use the prime-degree case of Capelli's binomial criterion [10, Chapter VI, Section 9]: over a characteristic-zero field $K$, the polynomial $x^p - a$, with $p$ prime, is irreducible unless $a \in K^p$.

**Proposition 15.8** (Prime-by-prime Kummer trigger)**.** *Consider selected orientations of packages with primitive shapes $(A, B)$ and $(C, D)$, scaling factors $r, s$, and a common root $\alpha$. Put*

$$g = \gcd(r,s), \quad r_0 = r/g, \quad s_0 = s/g, \quad w = \alpha^g, \quad \mu = \min(A+B, C+D) - 2.$$

*Let $\beta = w^{r_0}$ be a root of an irreducible factor $h \mid Q_{A,B}$ of degree $e$. For every prime $p \mid r_0$, at least one of the following holds:*

*(i)* $pe \leq \mu$;

*(ii)* $\beta \in \mathbb{Q}(\beta)^p$.

*If $pe > \mu$, the second alternative holds and hence all applicable conclusions of Corollary 15.5, Theorem 15.6, and Theorem 14.5 apply.*

*Proof.* The minimal polynomial $k_w$ is noncyclotomic because $|w| = |\alpha|^g \neq 1$. It divides $Q_{A,B}(x^{r_0})$ and $Q_{C,D}(x^{s_0})$, whose exponent-pair gcds are coprime. Theorem 14.2 gives

$$[\mathbb{Q}(w) : \mathbb{Q}] \leq \mu.$$

Fix $p \mid r_0$, put $u = w^{r_0/p}$, and set $K = \mathbb{Q}(\beta)$. If $\beta \notin K^p$, Capelli's criterion makes $x^p - \beta$ irreducible over $K$, so

$$[K(u) : K] = p.$$

Since $K(u) \subseteq \mathbb{Q}(w)$, it follows that $pe \leq [\mathbb{Q}(w) : \mathbb{Q}] \leq \mu$. The contrapositive gives the second alternative when $pe > \mu$. □

**Remark 15.9.** The former condition $2e > \mu$ is the special case in which the internal-power alternative is forced simultaneously for every prime divisor of $r_0$. The prime-by-prime form is stronger when the relative scaling exponent has a large prime divisor.

### 15.3 Role assignments and exact factor partitions

**Proposition 15.10** (Odd-prime degree signatures)**.** *Suppose a reciprocal irreducible orbit of degree $e$ occurs in primitive packages of totals $N$ and $M$. Put*

$$\rho_{\text{odd}} = \operatorname{lcm}\bigl(\operatorname{rad} N, \operatorname{rad} M\bigr)_{\text{odd}}.$$

*Then there is a unique factorization*

$$\rho_{\text{odd}} = UV, \qquad (U, V) = 1, \qquad U \mid e, \qquad V \mid e + 2, \tag{15.7}$$

*obtained by assigning each odd prime divisor of $\rho_{\text{odd}}$ to exactly one of the two roles $e \equiv 0$ or $e \equiv -2$. If $2 \mid NM$, then $e$ is even.*

*Proof.* For every odd $q \mid \rho_{\text{odd}}$, Theorem 15.1 says that $q \mid e(e + 2)$. Since $\gcd(e, e + 2) \mid 2$, the prime $q$ divides exactly one of the two integers. Multiplying the primes in their two roles gives (15.7). The assertion at 2 is the parity part of the same theorem. □

**Corollary 15.11** (Exact factor-partition signature)**.** *Let $N = qk$, where $q$ is an odd prime, and let the irreducible factor degrees of a primitive $Q_{A,B}$ be $e_1, \dots, e_t$. Write uniquely*

$$e_i = qa_i - 2\varepsilon_i, \qquad a_i \geq 1, \qquad \varepsilon_i \in \{0, 1\},$$

*and put $t_q = \sum_i \varepsilon_i$. Then for an integer $j \geq 0$,*

$$t_q = 1 + qj, \qquad \sum_i a_i = k + 2j, \qquad (q - 2)j \leq k - 1. \tag{15.8}$$

*In particular, if $q > k + 1$, then $j = 0$: exactly one factor has the $-2$-role at $q$, and the total number of irreducible factors is at most $k = N/q$. If $N = 2q$ with $q \geq 5$, the $q$-signature alone would force any reducible factorization to have two factors of degrees $q - 2$ and $q$.*

*Proof.* Summing the factor degrees gives

$$q \sum_i a_i - 2t_q = qk - 2.$$

Reduction modulo $q$ yields $t_q \equiv 1 \pmod{q}$, so $t_q = 1 + qj$, and substitution gives $\sum_i a_i = k + 2j$. Since each index counted by $t_q$ contributes at least one to $\sum_i a_i$,

$$1 + qj = t_q \le \sum_i a_i = k + 2j,$$

which is the final inequality in (15.8). If $q > k + 1$, then $q - 2 > k - 1$, forcing $j = 0$. For $k = 2$, reducibility forces two positive $a_i$'s summing to two, and exactly one $\varepsilon_i$ is one; the degrees are therefore $q - 2$ and $q$. $\square$

**Corollary 15.12** (Prime and twice-prime totals)**.** *Let $A, B$ be coprime and unequal. If $A + B = q$, or if $A + B = 2q$, where $q$ is an odd prime, then $Q_{A,B}$ is irreducible over $\mathbb{Q}$. Consequently every positive integral lift of such a primitive shape is globally isolated.*

*Proof.* If the total is $q$, Theorem 15.1 says that a factor degree $e$, with $1 \le e \le q - 2$, is congruent to 0 or $-2 \pmod{q}$. The only positive possibility in this range is $e = q - 2$, the full degree.

Now let the total be $2q$. The prime 2 in the radical sieve makes every factor degree even. The $q$-congruence shows that a proper factor degree in the range $1 \le e < 2q - 2$ can only be $q - 2$ or $q$, both odd. Thus no proper factor exists. Global isolation of all lifts follows from Theorem 14.9. $\square$

The following role rigidity is useful when a common factor is much smaller than a prime in the shared additive-triple support.

**Proposition 15.13** (Large-prime endpoint-role rigidity)**.** *Let an irreducible polynomial $h$ of degree $e$ divide selected primitive orientations $Q_{A,B}$ and $Q_{C,D}$. Let $p \ge 5$ be prime with $p > e + 2$, and suppose $p^\kappa \| B$. Then $p \nmid C + D$, and necessarily $p^\lambda \| D$ for some $\lambda \ge 1$. Moreover*

$$\frac{\kappa}{A} = \frac{\lambda}{C}, \qquad \frac{A}{\gcd(A, \kappa)} \mid e, \qquad \frac{C}{\gcd(C, \lambda)} \mid e. \tag{15.9}$$

*In particular, $\kappa = \lambda = 1$ is impossible for two distinct packages.*

*Proof.* Support compatibility, Corollary 11.5, forces $p \mid CD(C + D)$. It cannot divide $C + D$, because the radical sieve would give $e \equiv 0$ or $-2 \pmod{p}$, impossible when $0 < e < p - 2$.

At an endpoint prime, every unit-cluster side has reduced slope denominator at least $p - 2 > e$. A local factor of the degree-$e$ polynomial $h$ cannot contain such a root. Hence every root of $h$ belongs to the nonunit block. In $Q_{A,B}$ these valuations are positive and equal to $\kappa/A$; therefore the second orientation must also put $p$ in its constant endpoint, namely $D$, rather than in its leading endpoint $C$, whose nonunit valuations have the opposite sign. Equality of the valuations gives the first relation in (15.9). Integrality of the valuation of the norm gives the two denominator divisibilities.

If $\kappa = \lambda = 1$, then $A \mid e$ and $C \mid e$. The two nonunit blocks also give $e \le A, C$, so $A = C = e$, contradicting the disjointness of the two additive triples. $\square$

The merged 2-adic slope is therefore no longer an unresolved local case. The remaining bad-endpoint room is precise: an internal $p$-th power must lie entirely in a nonunit endpoint block and requires an endpoint valuation exponent at least $p$, together with the divisibilities in (15.5)–(15.6).

## 16 Reciprocal resultants as a supplementary sieve

The reciprocal resultant gives a compact numerical check on proposed factor sharing within a single package. It is not strong enough to prove package coprimality by itself, but it supplies useful endpoint information and an independent consistency test.

For primitive $a, b$, put $[k] = 1 + x + \cdots + x^{k-1}$. Reciprocal polynomials are normalized as in Section 4. One has

$$Q_{a,b}(x) + Q_{b,a}(x) = (a+b)[a](x)[b](x). \tag{16.1}$$

**Proposition 16.1** (Exact reciprocal resultant)**.** *If* $\gcd(a, b) = 1$*, then*

$$|\mathrm{Res}(Q_{a,b}, Q_{b,a})| = a^{a-2} b^{b-2} (a+b)^{a+b-2}, \tag{16.2}$$

*where a factor* $1^{-1}$ *is interpreted as* 1*.*

*Proof.* At a root $\alpha$ of $Q_{a,b}$, equation (16.1) gives

$$Q_{b,a}(\alpha) = (a+b)[a](\alpha)[b](\alpha).$$

At a nontrivial $a$-th root of unity $\zeta$,

$$Q_{a,b}(\zeta) = a\frac{\zeta^b - 1}{(\zeta - 1)^2}.$$

Because multiplication by $b$ permutes the nontrivial $a$-th roots,

$$|\,\mathrm{Res}(Q_{a,b}, [a])| = a^{a-2}.$$

Similarly, $|\,\mathrm{Res}(Q_{a,b}, [b])| = b^{b-2}$. More explicitly,

$$\begin{aligned} \mathrm{Res}(Q_{a,b}, Q_{b,a}) &= \mathrm{Res}(Q_{a,b}, (a+b)[a][b] - Q_{a,b}) \\ &= (a+b)^{a+b-2}\,\mathrm{Res}(Q_{a,b}, [a])\,\mathrm{Res}(Q_{a,b}, [b]). \end{aligned}$$

The preceding two evaluations give the formula. □

If $a = gA$, $b = gB$, $\gcd(A, B) = 1$, then composition and scaling give

$$|\mathrm{Res}(Q_{a,b}, Q_{b,a})| = g^{2g(A+B-2)} \left(A^{A-2} B^{B-2} (A+B)^{A+B-2}\right)^g. \tag{16.3}$$

These formulas are useful as a sieve for a proposed common irreducible factor, but cannot by themselves prove coprimality: a nonreciprocal irreducible polynomial may have reciprocal resultant 1.

## 17 Sparse Wronskians, explicit absolute boundedness, and arithmetic sieves

The proof now turns the structural restrictions into absolute bounds. Sparse Wronskians, torus-intersection rigidity, height estimates, and endpoint divisibility together force any normalized counterexample into an explicit finite range.

Let

$$W_{m,n;p,q}(x) = H'_{m,n}(x) H_{p,q}(x) - H_{m,n}(x) H'_{p,q}(x).$$

**Proposition 17.1** (Sparse Wronskian). *If a noncyclotomic irreducible polynomial $h$ divides both $H_{m,n}$ and $H_{p,q}$, then*

$$h(x)^2 \mid \frac{W_{m,n;p,q}(x)}{(x-1)^4}. \tag{17.1}$$

*Moreover the numerator has at most eight monomials before equal exponents are merged:*

$$\begin{aligned} W_{m,n;p,q}(x) = {} & mp(n-q)x^{n+q-1} + mq(p-n)x^{n+p-1} \\ & + np(q-m)x^{m+q-1} + nq(m-p)x^{m+p-1} \\ & + mn(q-p)(x^{n-1} - x^{m-1}) \\ & - pq(n-m)(x^{q-1} - x^{p-1}). \end{aligned} \tag{17.2}$$

*If the ordered edges are distinct, then $W_{m,n;p,q} \not\equiv 0$.*

*Proof.* Write $H_{m,n} = (x-1)^2 K_1$ and $H_{p,q} = (x-1)^2 K_2$. Then

$$W_{m,n;p,q} = (x-1)^4 (K_1' K_2 - K_1 K_2').$$

If $K_1 = hA$ and $K_2 = hB$, the terms containing $h'$ cancel:

$$K_1' K_2 - K_1 K_2' = h^2 (A'B - AB').$$

This proves (17.1), and direct expansion gives (17.2). If the Wronskian vanished identically, then $(H_{m,n}/H_{p,q})' = 0$, so the two trinomials would be scalar multiples. Comparison of their three nonzero supports $\{0, m, n\}$ and $\{0, p, q\}$ would force $(m, n) = (p, q)$. □

The Wronskian supplies a square-factor obstruction for each fixed exponent tuple. A uniform structural reduction comes from applying the low-height sparse-gcd theorem of Amoroso–Sombra–Zannier directly to the two collision trinomials. We first make the two geometric inputs completely formal.

**Lemma 17.2** (Four-unit divisor rigidity). *Let $C$ be a smooth projective curve over an algebraically closed field $K$ of characteristic zero, and let $u, v \in K(C)$ be nonconstant. If*

$$\dim_{\mathbb{Q}} \operatorname{span}_{\mathbb{Q}} \{\operatorname{div}(u), \operatorname{div}(u-1), \operatorname{div}(v), \operatorname{div}(v-1)\} \le 2,$$

*then*

$$v \in \left\{ u, 1-u, u^{-1}, (1-u)^{-1}, \frac{u}{u-1}, \frac{u-1}{u} \right\}.$$

*Proof.* Put

$$\Delta_0 = \operatorname{div}(u), \quad \Delta_1 = \operatorname{div}(u-1), \quad E_0 = \operatorname{div}(v), \quad E_1 = \operatorname{div}(v-1).$$

A nonconstant function on a smooth projective curve is surjective onto $\mathbb{P}^1$. Hence the fibres of $u$ over $0, 1, \infty$ are all nonempty. The divisors $\Delta_0, \Delta_1$ are linearly independent: at a point over $u = 0$, only $\Delta_0$ has positive valuation, and at a point over $u = 1$, only $\Delta_1$ does. Thus

$$E_0 = a\Delta_0 + b\Delta_1, \qquad E_1 = c\Delta_0 + d\Delta_1$$

for rational $a, b, c, d$. In particular, the zero, one, and pole fibres of $v$ are supported over the three special fibres of $u$. Moreover the role is uniform on each such fibre. Indeed, at a point $P$ over $u = 0$,

$$(\operatorname{ord}_P(v), \operatorname{ord}_P(v-1)) = \operatorname{ord}_P(u)(a, c),$$

while over $u = 1$ it is $\operatorname{ord}_P(u-1)(b,d)$, and over $u = \infty$ it is $-\operatorname{ord}_P(u^{-1})(a+b, c+d)$. The coefficient pair is independent of the chosen point in that fibre.

Because $v - (v-1) = 1$, the normalized valuation pair on each special fibre is therefore uniformly one of

$$(r,0), \quad (0,r), \quad (-r,-r) \quad (r>0), \qquad \text{or} \qquad (0,0).$$

Since the nonconstant function $v$ assumes all three values $0, 1, \infty$, the three nonneutral roles occur, and they occur bijectively on the three special fibres of $u$. Replacing $v$ by one of the six Möbius transformations permuting $\{0, 1, \infty\}$ preserves the same two-dimensional divisor span. We may therefore arrange that the zero, one, and pole fibres of $v$ lie respectively over those of $u$. Then, for positive rationals $r_0, r_1$,

$$\operatorname{div}(v) = r_0 \operatorname{div}(u), \qquad \operatorname{div}(v-1) = r_1 \operatorname{div}(u-1).$$

At the common pole fibre, $v$ and $v-1$ have the same negative valuation, so $r_0 = r_1 =: r$.

Write $r = A/B$ with $A, B$ positive coprime integers. The divisor equalities give constants $\kappa, \lambda \neq 0$ such that

$$v^B = \kappa u^A, \qquad (v-1)^B = \lambda(u-1)^A.$$

Logarithmic differentiation gives

$$B\frac{\mathrm{d}v}{v} = A\frac{\mathrm{d}u}{u}, \qquad B\frac{\mathrm{d}v}{v-1} = A\frac{\mathrm{d}u}{u-1}.$$

In characteristic zero, $\mathrm{d}u \not\equiv 0$, and hence also $\mathrm{d}v \not\equiv 0$. Comparing the two identities gives

$$\frac{v}{u} = \frac{v-1}{u-1},$$

so $v = u$. Undoing the initial Möbius permutation gives the six possibilities. □

**Proposition 17.3** (Rank-two torus-intersection rigidity). *Let $0 < m < n$, $0 < p < q$, assume*

$$\gcd\bigl(\gcd(m,n), \gcd(p,q)\bigr) = 1,$$

*and suppose the two unordered packages are distinct. Put*

$$X = V\bigl((n-m) - nx_1 + mx_2,\ (q-p) - qx_3 + px_4\bigr) \subset \mathbb{G}_m^4.$$

*There is no geometric irreducible curve contained in the intersection of $X$ with a two-dimensional subtorus and containing a point*

$$(\xi^m, \xi^n, \xi^p, \xi^q)$$

*for which $\xi \notin \mu_\infty$ is a common root of $H_{m,n}$ and $H_{p,q}$.*

*Proof.* Base-change to an algebraic closure and work on the smooth projective model of the normalization of such a curve $C$. Introduce

$$u = \frac{n}{n-m}x_1, \qquad u - 1 = \frac{m}{n-m}x_2,$$

$$v = \frac{q}{q-p}x_3, \qquad v - 1 = \frac{p}{q-p}x_4.$$

Suppose first that one projection of $C$ to a punctured factor line is constant. The other projection is then nonconstant and hence dominant. A monomial $x_3^a x_4^b$ constant on the second line must have $a = b = 0$: parameterize the line by

$$x_3 = t, \qquad x_4 = \frac{qt - (q-p)}{p},$$

and compare the valuations at $t = 0$, $t = (q-p)/q$, and infinity. Thus, if the first projection is constant, the two independent characters of the containing subtorus have zero last two coordinates. They remain independent in the first two coordinates, so the fixed point in $\mathbb{G}_m^2$ lies in a finite subgroup and is torsion. At the specified point, both $\xi^m$ and $\xi^n$ are roots of unity, forcing $\xi$ to be a root of unity. This is a contradiction. The other constant projection is symmetric.

Both $u$ and $v$ are therefore nonconstant. The annihilator lattice of the two-dimensional subtorus has rank two. Its two independent characters lie in the kernel of the divisor map

$$\mathbb{Q}^4 \longrightarrow \mathrm{Div}(C) \otimes \mathbb{Q}$$

sending the standard basis to $\mathrm{div}(u), \mathrm{div}(u-1), \mathrm{div}(v), \mathrm{div}(v-1)$. Hence the four divisors span a space of dimension at most two. Lemma 17.2 shows that $v = \sigma(u)$ for one of its six Möbius transformations.

At the point coming from $\xi$, this gives two binomial equations

$$\xi^{p-r_\sigma} = c_1, \qquad \xi^{q-s_\sigma} = c_2, \qquad c_1, c_2 \in \mathbb{Q}^\times,$$

where

| $\sigma(u)$ | $r_\sigma$ | $s_\sigma$ |
|---|---|---|
| $u$ | $m$ | $n$ |
| $1-u$ | $n$ | $m$ |
| $u^{-1}$ | $-m$ | $n-m$ |
| $(1-u)^{-1}$ | $-n$ | $m-n$ |
| $u/(u-1)$ | $m-n$ | $-n$ |
| $(u-1)/u$ | $n-m$ | $-m$ |

The two exponent differences cannot both vanish unless the ordered edges, and hence the packages, agree. Put

$$d = \gcd(|p - r_\sigma|, |q - s_\sigma|) > 0.$$

Bézout gives $\xi^d = c \in \mathbb{Q}^\times$. If a prime divided both $d$ and $\gcd(m,n)$, it would divide $r_\sigma, s_\sigma, p, q$, contrary to the coprimality hypothesis. Thus

$$\gcd(d, \gcd(m,n)) = 1.$$

The minimal polynomial of $\xi$ divides both $H_{m,n}$ and $x^d - c$. Lemma 14.1 gives degree at most one, and Theorem 10.1 then permits only the two orientations of the single package $\{1,2\}$. Two distinct packages are impossible. ☐

We now state the sparse-gcd handoff in precisely the form used below. For a torus homomorphism $\psi$, write $\psi^\#$ for pullback on Laurent polynomials.

**Proposition 17.4** (Specialized Amoroso–Sombra–Zannier output)**.** *Let*

$$\varphi(t) = (t^m, t^n, t^p, t^q), \qquad D = \max\{m, n, p, q\},$$

*where* $\gcd(m,n,p,q)=1$*, and let* $\xi \notin \mu_\infty$ *be a common root of* $H_{m,n}$ *and* $H_{p,q}$*. Put*

$$L_1 = (n-m) - nx_1 + mx_2, \qquad L_2 = (q-p) - qx_3 + px_4.$$

*Let* $B_4 = B'(4,1)$ *be the effective constant in Theorem 2.6 of [2]. If*

$$D^{1/6} > B_4(1 + \log(2D)), \tag{17.3}$$

*then there are* $0 \le k \le 3$*, an injective torus homomorphism*

$$\psi : \mathbb{G}_m^{4-k} \longrightarrow \mathbb{G}_m^4$$

*of size at most* $B_4$*, and* $\varphi_1 : \mathbb{G}_m \to \mathbb{G}_m^{4-k}$*, such that* $\varphi = \psi \circ \varphi_1$*. Put*

$$F_i = \psi^{\#} L_i, \qquad y_0 = \varphi_1(\xi).$$

*Either both* $F_i$ *vanish identically, or, with*

$$G = \gcd(F_1, F_2), \qquad g = \varphi_1^{\#} G,$$

*one has* $G(y_0) = 0$*.*

*Proof.* The coefficient height is at most $\log(2D)$, and the exponent vector is primitive, so Theorem 2.6 applies under (17.3). It gives the displayed factorization and says that $g \mid \gcd(H_{m,n}, H_{p,q})$. Moreover, a root of the quotient of this gcd by $g$ is either torsion or is a common root of the two proper-subsum equations obtained by retaining a nonempty proper subset $\Lambda \subset \{1,2,3,4\}$ of the four nonconstant monomials, while retaining the constant terms.

The point $\xi$ is not torsion. Let $I = \Lambda \cap \{1,2\}$. If $I$ is empty, the first partial equation is the nonzero constant $n-m$. If $I = \{1\}$, it is $(n-m) - n\xi^m = 0$; subtracting it from the full first equation gives $m\xi^n = 0$, impossible. If $I = \{2\}$, subtraction gives $-n\xi^m = 0$, also impossible. Thus $I = \{1,2\}$. The second equation similarly forces $\Lambda \cap \{3,4\} = \{3,4\}$. Hence $\Lambda$ would be the full set, contradicting properness. Therefore $g(\xi) = 0$, which is exactly $G(y_0) = 0$. The possibility that both pulled-back forms vanish identically is recorded separately. □

**Theorem 17.5** (Effective absolute boundedness of normalized counterexamples)**.** *There is an effectively computable absolute constant* $D_0$ *such that a normalized four-index counterexample*

$$0 < m < n, \qquad 0 < p < q, \qquad \gcd(m,n,p,q) = 1$$

*must satisfy*

$$\max\{m,n,p,q\} \le D_0.$$

*More explicitly, with* $B_4$ *as in Proposition 17.4, define*

$$D_{\mathrm{ASZ}} = \max\Big(\{1\} \cup \Big\{D \in \mathbb{N} : D^{1/6} \le B_4(1 + \log(2D))\Big\}\Big), \qquad D_0 = \max\{\lceil B_4 \rceil, D_{\mathrm{ASZ}}\}. \tag{17.4}$$

*Then this* $D_0$ *has the stated property.*

*Proof.* Assume $D > D_0$ and use Proposition 17.4. The two pulled-back forms cannot both vanish identically. Indeed, if

$$(n-m) - n\chi_1 + m\chi_2 = 0$$

as a Laurent polynomial in characters of a positive-dimensional torus, linear independence of distinct characters forces $\chi_1 = \chi_2 = 1$. The second form similarly forces $\chi_3 = \chi_4 = 1$, making $\psi$ trivial, contrary to injectivity. Thus $G(y_0) = 0$. Choose a geometric irreducible factor $P$ of $G$ vanishing at $y_0$; if exactly one $F_i$ is zero, take $P$ from the other one.

If $k = 3$, write the primitive exponent column of $\psi$ as $B = (b_1, b_2, b_3, b_4)^\mathsf{T}$, so

$$(m, n, p, q)^\mathsf{T} = B\theta.$$

Injectivity gives $\gcd(b_1, b_2, b_3, b_4) = 1$, while normalization gives $\gcd(m, n, p, q) = 1$. Hence $|\theta| = 1$ and $D = \|B\|_\infty \leq B_4$, contradicting $D > D_0$.

If $k = 0$, the injective endomorphism $\psi$ of a torus of the same dimension is an automorphism. Pullback therefore preserves coprimality, whereas $L_1, L_2$ are coprime Laurent linear forms. Thus $G$ is a unit, contradicting $G(y_0) = 0$.

Suppose $k = 1$. The hypersurface $V(P) \subset \mathbb{G}_m^3$ is a geometric irreducible surface. An injective homomorphism of tori is a closed immersion, so $\psi(V(P))$ is a closed irreducible surface contained in

$$X = V(L_1, L_2),$$

the irreducible product of two punctured lines. Since $\dim X = 2$, the image must equal $X$, placing $X$ inside the proper subtorus $\operatorname{im} \psi$.

This is impossible even for a mixed monomial relation. If

$$x_1^a x_2^b x_3^c x_4^d$$

were constant on the product, fixing the second factor would make $x_1^a x_2^b$ constant on the first line, and fixing the first would do the same for $x_3^c x_4^d$ on the second. Parameterizing the first line by

$$x_1 = t, \qquad x_2 = \frac{nt - (n - m)}{m}$$

and comparing the valuations at $0, (n - m)/n, \infty$ gives $a = b = 0$; symmetrically $c = d = 0$. Thus $X$ lies in no proper subtorus.

Finally let $k = 2$. The geometric curve $V(P) \subset \mathbb{G}_m^2$ has closed image contained in $X \cap \operatorname{im} \psi$ and passing through the specified point. The global normalization gives

$$\gcd\big(\gcd(m, n), \gcd(p, q)\big) = 1.$$

Proposition 17.3 excludes this curve. Every possible value of $k$ is contradictory, proving the bound. $\square$

**Corollary 17.6** (Finite-template reduction)**.** *Every counterexample is a simultaneous positive integral dilation of one of finitely many normalized configurations. In particular, the full package coprimality conjecture is reduced to an effective finite exact computation.*

*Proof.* Divide the four edge endpoints by their common gcd and replace the common root $z$ by the corresponding power, as in (2.5). The normalized configuration is bounded by Theorem 17.5. Conversely, the original configuration is its simultaneous dilation. $\square$

**Remark 17.7** (The symbolic ASZ threshold)**.** The constant $B_4$ is effective in the cited theorem, but no numerical value is supplied there. Thus Theorem 17.5, by itself, does not furnish a practical endpoint. The direct argument in the next subsection bypasses $B_4$ and produces an explicit numerical bound.

## 17.1 A direct numerical bound from height, geometry of numbers, and Smyth's theorem

The preceding Amoroso–Sombra–Zannier argument proves effective absolute boundedness without displaying a numerical constant. In the present four-variable situation one can obtain a substantially more concrete bound by combining an elementary height estimate with a short rank-two subtorus. We write $\mathrm{h}(\alpha)$ for the absolute logarithmic Weil height and $M(f)$ for the Mahler measure of a primitive polynomial $f \in \mathbb{Z}[x]$.

**Lemma 17.8** (Height of a collision root)**.** *Let $0 < u < v$, and suppose $H_{u,v}(\alpha) = 0$. Then*

$$(v-u)\,\mathrm{h}(\alpha) \le \log(2v), \qquad u\,\mathrm{h}(\alpha) \le \log(2v). \tag{17.5}$$

*Consequently,*

$$v\,\mathrm{h}(\alpha) \le 2\log(2v). \tag{17.6}$$

*Proof.* We use the elementary affine-height inequality

$$\mathrm{h}\Big(\frac{a\beta+b}{c}\Big) \le \mathrm{h}(\beta) + \log\!\big(2\max\{|a|,|b|,|c|\}\big), \qquad a,b,c \in \mathbb{Z},\ c \ne 0.$$

It follows place by place from the triangle inequality, with the factor two needed only at the archimedean places. The collision equation gives

$$\alpha^v = \frac{v\alpha^u - (v-u)}{u},$$

and hence

$$v\,\mathrm{h}(\alpha) \le u\,\mathrm{h}(\alpha) + \log(2v),$$

which is the first inequality in (17.5).

The reciprocal identity

$$x^v H_{v-u,v}(x^{-1}) = H_{u,v}(x)$$

shows that $\alpha^{-1}$ is a root of $H_{v-u,v}$. Applying the first inequality to that edge and using $\mathrm{h}(\alpha^{-1}) = \mathrm{h}(\alpha)$ gives the second inequality. Adding them proves (17.6). □

For a Euclidean lattice $\Lambda$, write $\det\Lambda$ for its covolume in its real span.

**Lemma 17.9** (Degree of a rank-two subtorus)**.** *Let $R \subset \mathbb{Z}^4$ be a saturated rank-two lattice, and let $T \subset \mathbb{G}_m^4$ be the two-dimensional subtorus annihilated by $R$. Under the standard embedding*

$$\mathbb{G}_m^4 \longrightarrow \mathbb{P}^4, \qquad (x_1,x_2,x_3,x_4) \longmapsto [1 : x_1 : x_2 : x_3 : x_4],$$

*the projective closure $\overline{T}$ satisfies*

$$\deg\overline{T} \le 2\det R. \tag{17.7}$$

*Proof.* Put $M = \mathbb{Z}^4/R$. Saturation makes $M$ a free rank-two lattice. Choose a lattice basis of $M$, and let $u_1,\dots,u_4 \in \mathbb{Z}^2$ be the images of the four standard basis vectors. With

$$\mathcal{A} = \{0, u_1, u_2, u_3, u_4\}, \qquad P = \mathrm{conv}(\mathcal{A}),$$

the closure $\overline{T}$ is the projective toric surface $X_{\mathcal{A}}$. The vectors $u_i$ generate $M$, so there is no lattice-index correction. The normalization of $X_{\mathcal{A}}$ is the normal toric surface associated with $P$, and finite birational normalization preserves the hyperplane degree. Hence the standard degree–volume theorem gives

$$\deg \overline{T} = \mathrm{Vol}_M(P),$$

where $\mathrm{Vol}_M$ is normalized lattice area: it is twice ordinary Euclidean area in coordinates in which a fundamental parallelogram of $M$ has area one [6, Theorems 3.A.5 and 13.4.3].

Let $U$ be the $2 \times 4$ integer matrix with columns $u_i$, and let $S \subset \mathbb{Z}^4$ be its row lattice. The quotient map $U : \mathbb{Z}^4 \to \mathbb{Z}^2$ is surjective and has kernel $R$. Thus $S$ is primitive; since its real span is $R^\perp$,

$$S = R^\perp \cap \mathbb{Z}^4.$$

For complementary primitive rank-two lattices in the unimodular lattice $\mathbb{Z}^4$, the covolumes agree. Concretely, the Euclidean Hodge star sends a primitive generator of $\bigwedge^2 R$ to a primitive generator of $\bigwedge^2 S$ and preserves norm. Hence $\det S = \det R$. The two rows of $U$ form a basis of $S$, so Cauchy–Binet gives

$$(\det R)^2 = (\det S)^2 = \det(UU^\mathsf{T}) = \sum_{1 \le i < j \le 4} \det(u_i, u_j)^2. \tag{17.8}$$

Order the vertices of $P$ cyclically. In the shoelace sum, every edge incident with the vertex 0 contributes zero. All other vertices belong to $\{u_1, \ldots, u_4\}$, so at most four nonzero determinant terms remain. Therefore Cauchy–Schwarz and (17.8) give

$$\deg \overline{T} \le 2 \left( \sum_{i<j} \det(u_i, u_j)^2 \right)^{1/2} = 2 \det R.$$

□

**Proposition 17.10** (A short rank-two subtorus)**.** *Let*

$$\boldsymbol{a} = (m, n, p, q) \in \mathbb{Z}^4$$

*be primitive, and put $D = \|\boldsymbol{a}\|_\infty$. There is a two-dimensional subtorus $T \subset \mathbb{G}_m^4$ containing the one-parameter subgroup*

$$t \longmapsto (t^m, t^n, t^p, t^q)$$

*and satisfying*

$$\deg \overline{T} < 4D^{2/3}. \tag{17.9}$$

*Proof.* Let

$$L = \boldsymbol{a}^\perp \cap \mathbb{Z}^4.$$

Primitivity of $\boldsymbol{a}$ means that the homomorphism $z \mapsto \langle \boldsymbol{a}, z \rangle$ maps $\mathbb{Z}^4$ onto $\mathbb{Z}$. Thus $L$ is saturated of rank three. Choose $c \in \mathbb{Z}^4$ with $\langle \boldsymbol{a}, c \rangle = 1$. A basis of $L$, together with $c$, is a basis of $\mathbb{Z}^4$; comparing base volume with the height $1/\|\boldsymbol{a}\|_2$ above $\boldsymbol{a}^\perp$ gives

$$\det L = \|\boldsymbol{a}\|_2 \le 2D.$$

Its Euclidean dual lattice $L^\vee$ has determinant $(\det L)^{-1}$. Minkowski's first theorem for the three-dimensional Euclidean ball [5, Chapter III, §2] provides a nonzero $y \in L^\vee$ with

$$\|y\|_2 \le \left( \frac{6}{\pi} \right)^{1/3} (\det L)^{-1/3}.$$

Divide by the content of $y$ in the free abelian group $L^\vee$. This can only decrease its norm, and it makes $y$ primitive, meaning that

$$L \longrightarrow \mathbb{Z}, \qquad b \longmapsto \langle b, y\rangle$$

is surjective. Set

$$R = \{b \in L : \langle b, y\rangle = 0\}.$$

Then $L/R \simeq \mathbb{Z}$, so $R$ is saturated in $L$. It is also saturated in $\mathbb{Z}^4$: in the exact sequence

$$0 \longrightarrow L/R \longrightarrow \mathbb{Z}^4/R \longrightarrow \mathbb{Z}^4/L \longrightarrow 0$$

both outer groups are torsion-free, hence so is $\mathbb{Z}^4/R$.

Choose a basis $b_1, b_2$ of $R$ and $b_3 \in L$ with $\langle b_3, y\rangle = 1$. The Euclidean height of $b_3$ above $R \otimes_{\mathbb{Z}} \mathbb{R}$ is $1/\|y\|_2$. Therefore

$$\det L = \frac{\det R}{\|y\|_2}, \qquad \det R = (\det L)\|y\|_2 \le \left(\frac{6}{\pi}\right)^{1/3} (2D)^{2/3}.$$

Let $T$ be the subtorus annihilated by $R$. Since $R \subset \boldsymbol{a}^\perp$, every character in $R$ is trivial on the displayed one-parameter subgroup; hence $T$ contains that subgroup. Lemma 17.9 gives

$$\deg \overline{T} \le 2\left(\frac{6}{\pi}\right)^{1/3} (2D)^{2/3} = 2\left(\frac{24}{\pi}\right)^{1/3} D^{2/3} < 4D^{2/3},$$

because $\pi > 3$. $\square$

**Proposition 17.11** (Degree of a normalized common-root field)**.** *Let*

$$0 < m < n, \qquad 0 < p < q, \qquad \gcd(m, n, p, q) = 1,$$

*let the two packages be distinct, and suppose that $\alpha \notin \mu_\infty$ is a common root of $H_{m,n}$ and $H_{p,q}$. Put*

$$D = \max\{m, n, p, q\}, \qquad d = [\mathbb{Q}(\alpha) : \mathbb{Q}].$$

*Then*

$$d < 4D^{2/3}. \tag{17.10}$$

*Proof.* Take the subtorus $T$ from Proposition 17.10. It contains

$$P_\alpha = (\alpha^m, \alpha^n, \alpha^p, \alpha^q).$$

The variety cut out in $\mathbb{G}_m^4$ by the two collision lines is the variety $X$ of Proposition 17.3. That proposition shows that $P_\alpha$ is isolated in $T \cap X$: a positive-dimensional local component would contain a geometric irreducible curve through the point. Since $P_\alpha$ lies in the open torus, a positive-dimensional component of $\overline{T} \cap \overline{X}$ through it would restrict near that point to one in $T \cap X$. Hence it is also an isolated projective intersection point.

Both $T$ and $X$, and therefore their projective closures, are defined over $\mathbb{Q}$. The closure $\overline{X}$ is the intersection of the two rational hyperplanes

$$(n-m)x_0 - nx_1 + mx_2 = 0, \qquad (q-p)x_0 - qx_3 + px_4 = 0$$

in $\mathbb{P}^4$, so $\deg \overline{X} = 1$. The generalized Bezout inequality for isolated components [7, §8.4] gives

$$\deg((\overline{T} \cap \overline{X})_{\mathrm{isol}}) \le \deg \overline{T} \deg \overline{X} = \deg \overline{T}.$$

Every Galois conjugate of $P_\alpha$ is again isolated and contributes positive intersection multiplicity. The closed point determined by this orbit therefore contributes at least $[\mathbb{Q}(P_\alpha) : \mathbb{Q}]$, whence

$$[\mathbb{Q}(P_\alpha) : \mathbb{Q}] \le \deg \overline{T} < 4D^{2/3}.$$

Finally, choose integers $r_1, r_2, r_3, r_4$ with $r_1 m + r_2 n + r_3 p + r_4 q = 1$. Since all four coordinates are nonzero,

$$\alpha = (\alpha^m)^{r_1} (\alpha^n)^{r_2} (\alpha^p)^{r_3} (\alpha^q)^{r_4}.$$

Thus $\mathbb{Q}(P_\alpha) = \mathbb{Q}(\alpha)$, proving (17.10). □

Let

$$\theta_0 = 1.3247179572447460259\ldots$$

be the plastic number, the real root greater than one of $T^3 - T - 1$.

**Proposition 17.12** (A uniform Mahler-measure gap). *Let $h \in \mathbb{Z}[x]$ be the primitive irreducible polynomial, with positive leading coefficient, of a noncyclotomic common root of two collision trinomials. Then $h$ is nonreciprocal and*

$$M(h) \ge \theta_0. \tag{17.11}$$

*Proof.* The polynomial $h$ divides an orientation $Q_{a,b}$. If it were self-reciprocal, it would also divide the reciprocal orientation $Q_{b,a}$, contradicting the nonzero reciprocal resultant (16.3). Thus $h$ is nonreciprocal.

If $h$ is monic, Smyth's theorem [12] gives $M(h) \ge \theta_0$. If it is not monic, its positive integral leading coefficient is at least two, and $M(h) \ge 2 > \theta_0$. □

**Theorem 17.13** (Explicit absolute bound for normalized counterexamples). *Every normalized four-index counterexample*

$$0 < m < n, \qquad 0 < p < q, \qquad \gcd(m, n, p, q) = 1$$

*satisfies*

$$\boxed{\max\{m, n, p, q\} \le 175\,394\,637.} \tag{17.12}$$

*Proof.* Let $\alpha$ be a common root, let $h$ be its primitive irreducible polynomial, let $d = \deg h$, and put $D = \max\{m, n, p, q\}$. One of the two collision edges has upper endpoint $D$. Lemma 17.8 gives

$$\mathrm{h}(\alpha) \le \frac{2\log(2D)}{D}. \tag{17.13}$$

By Proposition 17.11, $d < 4D^{2/3}$. Since $\log M(h) = d\,\mathrm{h}(\alpha)$, Proposition 17.12 yields

$$\log \theta_0 \le \log M(h) < \frac{8\log(2D)}{D^{1/3}}. \tag{17.14}$$

The function on the right is strictly decreasing for $D \ge 11$. Certified real-ball arithmetic gives

$$\frac{8\log(2 \cdot 175394637)}{175394637^{1/3}} > \log \theta_0,$$

whereas

$$\frac{8\log(2\cdot 175394638)}{175394638^{1/3}} < \log\theta_0.$$

Thus (17.14) is impossible for $D \ge 175394638$. The smaller values $D < 11$ are already below the asserted bound. $\square$

**Corollary 17.14** (The upper range is unit-normalized)**.** *In a normalized counterexample with*

$$D = \max\{m, n, p, q\} \ge 6\,816\,242,$$

*the primitive irreducible polynomial $h$ of a common root is monic and has constant coefficient $\pm 1$.*

*Proof.* The same decreasing function satisfies

$$\frac{8\log(2\cdot 6816242)}{6816242^{1/3}} < \log 2.$$

Hence (17.13) and Proposition 17.11 give $M(h) < 2$. Both the absolute leading coefficient and the absolute constant coefficient of a primitive polynomial are at most its Mahler measure. They are nonzero integers, so both equal one. $\square$

**Remark 17.15.** Theorem 17.13 removes the practical indeterminacy in the constant $B_4$. The Amoroso–Sombra–Zannier route remains an independent structural proof of effective boundedness, while the direct argument gives a fully numerical endpoint. The two tiny numerical comparisons above can be certified by the real-ball script recorded in Section 18.

## 17.2 Moment, discriminant, and endpoint sieves

The explicit box is still large. The following additional restrictions are intended for the finite computation. They are extracted from the all-moment form of Borisov's cluster-count argument.

Let

$$h(x) = ux^e + c_{e-1}x^{e-1} + \cdots + c_1x + v \in \mathbb{Z}[x], \qquad u > 0,$$

be primitive and irreducible. For its roots $\beta$, counted with multiplicity, put

$$A_j(h) = \sum_{h(\beta)=0} (1-\beta)\beta^j \qquad (j \ge 0). \tag{17.15}$$

**Proposition 17.16** (All-moment congruence)**.** *Suppose $h$ divides two selected primitive orientations $Q_{A,B}$ and $Q_{C,D}$, where*

$$(A, B) = (C, D) = 1, \qquad N = A + B, \qquad M = C + D.$$

*If a prime $\ell$ divides $ABNCDM$ but does not divide $uv$, then*

$$A_j(h) \in \ell\mathbb{Z}_{(\ell)} \qquad \textit{for every } j \ge 0. \tag{17.16}$$

*Proof.* Use a package whose additive triple contains $\ell$. Since $\ell \nmid uv$, the leading and constant coefficients of $h$ are $\ell$-adic units. The Newton polygon of $h$ is therefore horizontal, so all roots of $h$ are $\ell$-adic units.

Borisov's cluster argument [1, Lemma 3.1 and Remark 3.5] shows that, at every nontrivial prime-to-$\ell$ root-of-unity center, the number of roots belonging to a rational factor is divisible by $\ell$.

Modulo $\ell$, the summand in (17.15) is constant on such a cluster, namely $(1-\zeta)\zeta^j$, so the cluster contribution vanishes. At the center 1, every summand already vanishes modulo $\ell$. If $\ell$ is an endpoint prime, the nonunit endpoint block is absent because every root of $h$ is a unit. These contributions exhaust the roots and prove (17.16). The argument also covers $\ell = 2$; the merged side still contributes an even number of roots to every rational factor, as in Proposition 15.4. □

Define the reciprocal coefficient polynomial

$$\widetilde{h}(t) = t^e h(t^{-1}) = u + c_{e-1}t + \cdots + c_1 t^{e-1} + vt^e.$$

**Corollary 17.17** (Differential moment congruence)**.** *Under the hypotheses of Proposition 17.16, put*

$$\mathcal{S} = \operatorname{rad}(ABNCDM), \qquad \mathcal{R} = \prod_{\substack{\ell \mid \mathcal{S} \\ \ell \nmid uv}} \ell.$$

*Then every coefficient of*

$$(1-t)\widetilde{h}'(t) + e\widetilde{h}(t) \tag{17.17}$$

*is divisible by $\mathcal{R}$. In particular,*

$$\mathcal{R} \mid eu + c_{e-1}, \qquad \mathcal{R} \mid ev + c_1. \tag{17.18}$$

*Proof.* For each prime in $\mathcal{R}$, the constant coefficient $\widetilde{h}(0) = u$ is a unit. The generating series of the moments is

$$\sum_{j\ge 0} A_j(h)t^j = e + (1-t)\frac{\widetilde{h}'(t)}{\widetilde{h}(t)}.$$

Proposition 17.16 makes the left side zero modulo the prime. Multiplication by $\widetilde{h}(t)$ gives (17.17), and the constant and $t^{e-1}$ coefficients give (17.18). □

**Proposition 17.18** (Hankel–discriminant divisibility)**.** *With the same notation,*

$$\boxed{\mathcal{R}^e \mid h(1)\operatorname{Disc}(h).} \tag{17.19}$$

*Consequently,*

$$\mathcal{R} \le 2eM(h)^{2-1/e}. \tag{17.20}$$

*Proof.* Form the $e \times e$ Hankel matrix

$$\mathcal{H} = (A_{i+j}(h))_{0\le i,j<e}.$$

Writing $w_k = 1 - \beta_k$, its Vandermonde factorization is

$$\mathcal{H} = V\ \operatorname{diag}(w_1,\dots,w_e)V^{\mathsf{T}}.$$

Hence, up to sign,

$$\det\mathcal{H} = \prod_k (1-\beta_k)\prod_{i<j}(\beta_i-\beta_j)^2 = \frac{h(1)\operatorname{Disc}(h)}{u^{2e-1}}. \tag{17.21}$$

Every entry is divisible by every prime in $\mathcal{R}$, while $u$ is a unit at those primes. Thus (17.19) follows.

The elementary estimates

$$|h(1)| \le 2^e M(h), \qquad |\operatorname{Disc}(h)| \le e^e M(h)^{2e-2}$$

(the second is Mahler's discriminant inequality [11]) imply

$$\mathcal{R}^e \le 2^e e^e M(h)^{2e-1}.$$

Taking $e$-th roots proves (17.20). $\square$

**Corollary 17.19** (Support bound for two colliding primitive packages)**.** *Under the same hypotheses,*

$$\mathcal{S} \le 2eM(h)^{4-1/e}. \tag{17.22}$$

*If $h$ is monic with constant coefficient $\pm 1$, then*

$$\mathcal{S} \le 2eM(h)^{2-1/e}. \tag{17.23}$$

*Proof.* The primes of $\mathcal{S}$ omitted from $\mathcal{R}$ divide $uv$. Since

$$\operatorname{rad}(uv) \le |uv| \le M(h)^2,$$

formula (17.20) gives (17.22). In the unit-normalized case no prime is omitted, so $\mathcal{S} = \mathcal{R}$, giving (17.23). $\square$

**Proposition 17.20** (Endpoint divisibilities and slope matching)**.** *Under the same hypotheses,*

$$|v|^A \mid B^e, \qquad u^B \mid A^e, \qquad |v|^C \mid D^e, \qquad u^D \mid C^e. \tag{17.24}$$

*Moreover, for every prime $\ell \mid u$,*

$$\frac{v_\ell(A)}{B} = \frac{v_\ell(C)}{D}, \tag{17.25}$$

*and, for every prime $\ell \mid v$,*

$$\frac{v_\ell(B)}{A} = \frac{v_\ell(D)}{C}. \tag{17.26}$$

*Proof.* Because $Q_{A,B}$ is primitive with leading coefficient $A$ and constant coefficient $B$, Gauss's lemma gives an integral complementary factor. Hence $u \mid A$, $|v| \mid B$, and $(u,v) = 1$. At a root $\beta$ of $h$,

$$\beta^A(N - A\beta^B) = B.$$

Taking the resultant with $h$ gives

$$\operatorname{Res}(h, N - Ax^B) = \pm \frac{B^e u^N}{v^A} \in \mathbb{Z}. \tag{17.27}$$

Since $(u,v) = 1$, this proves $|v|^A \mid B^e$. Applying the same argument to the reciprocal factor in $Q_{B,A}$ proves $u^B \mid A^e$. The second package gives the remaining two divisibilities.

Now let $\ell \mid u$. Then $\ell \mid A, C$ and $\ell \nmid v$. The product of the roots of $h$ has negative $\ell$-adic valuation, so at least one root is nonunit. Borisov's endpoint polygons say that every negative-valuation root of $Q_{A,B}$ has valuation $-v_\ell(A)/B$, and every such root of $Q_{C,D}$ has valuation $-v_\ell(C)/D$. The same root lies in both polynomials, proving (17.25). The positive-valuation endpoint description gives (17.26) in exactly the same way when $\ell \mid v$. $\square$

**Remark 17.21.** For a finite search, the support bound, the four endpoint divisibilities, and the slope equalities should be applied before any polynomial factorization. For a surviving exponent tuple, let $S \in \mathbb{Z}[x]$ be the primitive part of $W_{m,n;p,q}/(x-1)^4$. Proposition 17.1 then permits a cheap modular certificate: if an auxiliary prime does not divide the leading coefficient of $S$—equivalently, reduction preserves its degree—and $\overline{S}$ is square-free over the corresponding finite field, then $S$ is square-free over $\mathbb{Q}$, so no common irreducible factor can exist. The good-reduction condition is essential: at a prime where the degree drops, a square factor can disappear after reduction.

## 18 Exact upper-range elimination and reproducibility

This is the first decisive computational section. All filters used here are exact integer filters derived above; the computation does not rely on numerical root finding or on factoring the large collision polynomials.

Table 1 lists, for each computer-assisted result in Sections 18–21, the generator that produces its data and the independent script that re-checks it.

| Result | Generator $\to$ independent verifier |
|---|---|
| Upper range, Thm 18.9 | `archive/Qab9_upper_bundle` pipeline $\to$ `verify_upper_safe.py`; constants `certify_constants.py`, `certify_thresholds.sage` |
| Unit branch, Thm 20.2 | `enumerate_unit_*.cpp`, `pair_unit_*.cpp` $\to$ `verify_unit_branch_manifest.py`, `verify_modular_gcd_certificates.py`, `verify_irreducibility_certificates_direct.py` |
| Both-nonunit, Thm 21.1 | `enumerate_two_nonunit_packages.cpp`, `pair_two_nonunit.cpp` $\to$ `verify_two_nonunit.py` |
| One-nonunit $D \geq 16\,584$, Thm 21.2 | `enumerate_one_nonunit_packages.cpp`, `pair_one_nonunit.cpp` $\to$ `verify_one_nonunit.py` |
| Residual one-nonunit, Thm 21.4 | `enumerate_one_nonunit_residual_packages.cpp`, `pair_one_nonunit_residual.cpp` $\to$ `verify_one_nonunit_residual.py` |
| Analytic constants | `certify_qab12_constants.py` |

Table 1: Paper-to-code correspondence for the computer-assisted steps. Scripts live in `code/qab12` and data in `data/qab12`, except the upper-range pipeline in `archive/Qab9_upper_bundle`; reproduction commands are in Section 18.

Put

$$D_0 = 6\,816\,242, \qquad D_1 = 175\,394\,637.$$

This section gives the mathematical reductions and the exact computation which exclude every normalized counterexample with $D_0 \leq D \leq D_1$. Write the two unordered packages in primitive form as

$$\{rA, rB\}, \qquad \{sC, sE\}, \qquad \gcd(A, B) = \gcd(C, E) = 1,$$

put $N = A + B$, $M = C + E$, and order the shapes by $A < B$, $C < E$. The corresponding collision upper endpoints are $rN$ and $sM$, so

$$D = \max(rN, sM).$$

Normalization gives $\gcd(r, s) = 1$. Let $\alpha$ be a common root, let $h_\alpha$ be its primitive minimal polynomial, and put

$$d = [\mathbb{Q}(\alpha) : \mathbb{Q}], \qquad \beta = \alpha^r, \qquad \gamma = \alpha^s.$$

The elements $\beta$ and $\gamma$ are roots of selected reciprocal orientations of the two primitive shapes.

### 18.1 An exact uniform degree gap and small scales

**Lemma 18.1** (An algebraic certificate for the plastic number)**.** *Let $\theta_0$ be the real root greater than one of $T^3 - T - 1$. Then*

$$\theta_0^{70} = 62\,608\,681 + 109\,870\,576\theta_0 + 82\,938\,844\theta_0^2 > 2D_1. \tag{18.1}$$

*Proof.* Reduction of $T^{70}$ modulo $T^3 - T - 1$ gives the displayed identity. The polynomial $T^3 - T - 1$ is strictly increasing for $T \geq 1$, and

$$\left(\frac{33}{25}\right)^3 - \frac{33}{25} - 1 = -\frac{313}{15625} < 0.$$

Hence $\theta_0 > 33/25$. Substitution in the positive-coefficient right-hand side of (18.1) gives the exact rational lower bound

$$\theta_0^{70} > \frac{220\,094\,051\,941}{625} > 350\,789\,274 = 2D_1.$$

$\square$

**Proposition 18.2** (Uniform degree and scale bounds)**.** *For a normalized counterexample with $D \leq D_1$,*

$$d > \frac{D}{140}. \tag{18.2}$$

*In the primitive decomposition above,*

$$1 \leq r, s \leq 139. \tag{18.3}$$

*In particular, throughout the unit range $D \geq D_0$,*

$$d \geq \left\lfloor \frac{D_0}{140} \right\rfloor + 1 = 48\,688.$$

*Proof.* The height estimate (17.13) and the Mahler gap (17.11) give

$$d \geq \frac{D \log \theta_0}{2 \log(2D)}.$$

Lemma 18.1 gives $70 \log \theta_0 > \log(2D_1) \geq \log(2D)$, proving (18.2). The confluent degree bound, Theorem 14.2, gives

$$d \leq \min(N, M) - 2 < N, M.$$

Since $rN \leq D$ and $sM \leq D$, the inequalities $D/140 < d < N \leq D/r$ and $D/140 < d < M \leq D/s$ imply $r, s < 140$. $\square$

## 18.2 Full Kummer degree in the unit range

The prime-degree form of Capelli's criterion used earlier has the following standard composite-degree form: over a characteristic-zero field $K$, $X^n - a$ is irreducible if $a \notin K^p$ for every prime $p \mid n$, and, when $4 \mid n$, also $a \notin -4K^4$; see [10, Chapter VI, Section 9].

**Proposition 18.3** (Full Kummer degree)**.** *Assume $D_0 \le D \le D_1$. Let $e_1 = [\mathbb{Q}(\beta) : \mathbb{Q}]$ and $e_2 = [\mathbb{Q}(\gamma) : \mathbb{Q}]$. Then*

$$X^r - \beta \quad \textit{is irreducible over } \mathbb{Q}(\beta), \qquad X^s - \gamma \quad \textit{is irreducible over } \mathbb{Q}(\gamma).$$

*Consequently there is an integer $k \ge 1$ such that*

$$d = re_1 = se_2 = rsk, \qquad e_1 = sk, \qquad e_2 = rk. \tag{18.4}$$

*Proof.* Corollary 17.14 makes $\alpha$, and hence $\beta, \gamma$, algebraic units. Fix a prime $p \mid r$, put $K = \mathbb{Q}(\beta)$, and suppose first that $\beta \in K^p$.

If $p \mid N$, Corollary 15.5 gives an immediate contradiction. If $p \mid AB$, the proof of Theorem 15.6 shows that an internal $p$-th power factor must lie entirely in the nonunit endpoint block, again impossible for the unit $\beta$. The only remaining case is $p \nmid ABN$. Then $p$ is odd, and Theorem 14.5 gives $e_1 \le p - 2$. Since $\alpha$ satisfies $X^r - \beta$ over $K$,

$$d \le re_1 \le 139 \cdot 137 = 19\,043,$$

contrary to $d \ge 48\,688$. Thus $\beta \notin K^p$ for every prime $p \mid r$.

It remains to exclude Capelli's exceptional fourth-power case. If $4 \mid r$, then 2 occupies exactly one role in the primitive triple $(A, B, N)$. The local proofs of Corollary 15.5 and Theorem 15.6, including the residual splitting of the merged side, place every unit root, after an unramified base extension, in a packet for which

$$v_2(\beta - \zeta) = \frac{1}{\ell} < 1, \qquad [L(\beta) : L] \le \ell,$$

with $\zeta$ a root of unity of odd order and $\ell \ge 2$. Since

$$-\beta - \zeta = -((\beta - \zeta) + 2\zeta),$$

the two summands have unequal valuations and $v_2(-\beta - \zeta) = 1/\ell$. Lemma 14.3 applied to $-\beta$ shows that $-\beta$ is not a square in $K$. In particular $\beta \notin -4K^4$, because $\beta = -4\eta^4$ would give $-\beta = (2\eta^2)^2$. Capelli's criterion now proves the irreducibility of $X^r - \beta$. The proof for $X^s - \gamma$ is identical.

It follows that $d = re_1 = se_2$. Since $\gcd(r, s) = 1$, one has $s \mid e_1$ and $r \mid e_2$, giving (18.4). $\square$

## 18.3 One-package support and full-radical congruences

**Proposition 18.4** (One-package unit support bound)**.** *Let $f \in \mathbb{Z}[x]$ be a monic irreducible factor of a selected primitive orientation of shape $(A, B)$, let $e = \deg f$, and suppose $f(0) = \pm 1$. Then*

$$\mathrm{rad}(AB(A + B)) \le 2eM(f)^{2-1/e}. \tag{18.5}$$

*Proof.* For every prime dividing $AB(A+B)$, the proof of Proposition 17.16 uses only the one primitive package containing that prime. Since the leading and constant coefficients of $f$ are units, its endpoint nonunit block is absent. The same all-moment congruence, Hankel determinant, and Mahler discriminant estimate as in Proposition 17.18 therefore apply with $\mathcal{R} = \operatorname{rad}(AB(A+B))$, giving (18.5). ☐

**Corollary 18.5** (The exact support cap used by the computation)**.** *Under the hypotheses of Proposition 18.3,*

$$r\,\operatorname{rad}(ABN) < 4\,398\,937, \qquad s\,\operatorname{rad}(CEM) < 4\,398\,937. \tag{18.6}$$

*Proof.* Let $f_\beta$ be the minimal polynomial of $\beta$. From (18.4),

$$\log M(f_\beta) = e_1\,\mathrm{h}(\beta) = e_1 r\,\mathrm{h}(\alpha) = d\,\mathrm{h}(\alpha) = \log M(h_\alpha).$$

Since $\beta$ is noncyclotomic, $M(f_\beta) > 1$. By Propositions 17.11 and 18.4, and using $e_1 < 4D^{2/3}/r$ together with (17.13), we obtain

$$\begin{aligned} r\,\operatorname{rad}(ABN) &\le 2re_1 M(f_\beta)^{2-1/e_1} \\ &< 8D^{2/3}\exp\Bigl(\frac{16\log(2D)}{D^{1/3}}\Bigr) =: C(D). \end{aligned}$$

Indeed, $\log M(f_\beta) = \log M(h_\alpha)$ and $\log M(h_\alpha) < 8\log(2D)/D^{1/3}$. Writing $x = D^{1/3}$,

$$\frac{\mathrm{d}}{\mathrm{d}x}\log C(x^3) = \frac{2x + 48 - 16\log(2x^3)}{x^2} > 0$$

throughout $D_0 \le D \le D_1$: here $x > 189$ and $\log(2D) < 20$, so the numerator is greater than 106. Thus it is enough to bound $C(D_1)$. The exact rational certificate listed in Table 1 uses

$$\frac{559\,764\,608}{10^6} < D_1^{1/3} < \frac{559\,764\,609}{10^6}$$

and rational upper bounds from the atanh series for log and the Taylor series for exp. It proves

$$C(D_1) < 4\,398\,937.$$

The second package is symmetric. ☐

**Proposition 18.6** (Full-radical degree signature in the unit case)**.** *Under the hypotheses of Proposition 18.4, for every prime $p \mid AB(A+B)$,*

$$e \equiv 0 \quad \textit{or} \quad -2 \pmod{p}. \tag{18.7}$$

*Thus* $\operatorname{rad}(AB(A+B)) \mid e(e+2)$.

*Proof.* For a total prime this is Theorem 15.1. At an endpoint prime the unit hypothesis removes the endpoint nonunit block. Borisov's remaining root-of-unity clusters have contributions divisible by $p$, except that the center-1 packet contributes either 0 or $p-2$ modulo odd $p$. For $p = 2$, every packet contribution is even, including the residual-split merged side. This proves (18.7) in every role. ☐

### 18.4 Role-binomial degree bounds

**Lemma 18.7** (Reduction to a role binomial)**.** *Let $p$ divide exactly one member $R$ of the primitive triple $(A, B, N)$, and let $h_\alpha$ be monic with constant coefficient $\pm 1$. If $h_\alpha$ divides a selected orientation $Q_{A,B}(X^r)$ or $Q_{B,A}(X^r)$, then*

$$\overline{h_\alpha} \mid X^{rR} - 1 \quad \text{in } \mathbb{F}_p[X], \qquad d \le rR. \tag{18.8}$$

*If the same prime occupies role $S$ in the second primitive triple, then*

$$d \le \gcd(rR, sS).$$

*Proof.* For either orientation, direct reduction of the collision trinomial gives

$$H_{T,N}(X) \equiv cX^\nu (X^R - 1) \pmod p$$

for some $c \in \mathbb{F}_p^\times$, $\nu \ge 0$, where $T \in \{A, B\}$ is the selected endpoint. Hence

$$(X^r - 1)^2 Q_{T,N-T}(X^r) = H_{T,N}(X^r) \equiv cX^{r\nu}(X^{rR} - 1) \pmod p.$$

The monic unit polynomial $h_\alpha$ has degree-preserving reduction and is not divisible by $X$, proving the first assertion. If the prime appears in both shapes, use $\gcd(X^a - 1, X^b - 1) = X^{\gcd(a,b)} - 1$, valid in every characteristic. □

### 18.5 The same-shape Bezout bound

**Proposition 18.8** (Same primitive shape)**.** *Suppose distinct normalized packages have the same primitive shape $\{A, B\}$, with coprime scales $r, s$, and share a noncyclotomic irreducible factor of degree $d$. Then*

$$d \le 2rs. \tag{18.9}$$

*Proof.* After replacing the common root by its inverse if necessary, put $x = \alpha^r$ and choose $y = \alpha^s$ or $\alpha^{-s}$ so that both $x, y$ are roots of $Q_{A,B}$. According as the two selected orientations agree or disagree,

$$x^s = y^r \qquad \text{or} \qquad x^s y^r = 1.$$

Put $N = A + B$ and

$$u = \frac{A}{N} x^B, \qquad 1 - u = \frac{B}{N} x^{-A}, \qquad v = \frac{A}{N} y^B, \qquad 1 - v = \frac{B}{N} y^{-A}.$$

Because $\gcd(A, B) = 1$, the map $x \mapsto u$ is injective on the roots under consideration: $u$ determines both $x^B$ and $x^{-A}$, and an integral Bezout combination recovers $x$.

In the same-orientation case the point $(u, v)$ lies on

$$u^s = c_0 v^r, \qquad (1-u)^s = c_1 (1-v)^r;$$

in the opposite-orientation case it lies on

$$u^s v^r = c_0, \qquad (1-u)^s (1-v)^r = c_1,$$

with nonzero rational constants. In either case these are two divisors of bidegree $(s, r)$ on $\mathbb{P}^1 \times \mathbb{P}^1$, whose intersection number is $2rs$.

They have no common curve component. Indeed, if one of $u, v$ were constant on a component, either binomial equation would make the other constant as well, so the component would be a point. On the normalization of a genuine curve component, both $u, v$ are therefore nonconstant, and the two equations give two linear relations among

$$\Delta_0 = \mathrm{div}(u), \quad \Delta_1 = \mathrm{div}(1-u), \quad E_0 = \mathrm{div}(v), \quad E_1 = \mathrm{div}(1-v).$$

Lemma 17.2 forces $v$ to be one of the six transforms of $u$. In the basis $(\Delta_0, \Delta_1)$, their divisor pairs are

| $v$ | $E_0$ | $E_1$ |
|---|---|---|
| $u$ | $(1,0)$ | $(0,1)$ |
| $1-u$ | $(0,1)$ | $(1,0)$ |
| $u^{-1}$ | $(-1,0)$ | $(-1,1)$ |
| $(1-u)^{-1}$ | $(0,-1)$ | $(1,-1)$ |
| $u/(u-1)$ | $(1,-1)$ | $(0,-1)$ |
| $(u-1)/u$ | $(-1,1)$ | $(-1,0)$ |

The same-orientation relations are $s\Delta_i = rE_i$; the table permits only $v = u$ and $r = s$, which would give the same package. The opposite-orientation relations are $s\Delta_i + rE_i = 0$, and no row satisfies both. Thus the intersection is zero-dimensional.

The $d$ conjugates of $\alpha$ give $d$ distinct intersection points. Indeed equality of the corresponding $(u, v)$ recovers both $\alpha^r$ and $\alpha^{\pm s}$, and $\gcd(r, s) = 1$ recovers $\alpha$. Bezout's theorem now gives (18.9). □

## 18.6 The exact enumeration

The proof pipeline deliberately avoids constructing any large polynomial. Corollary 18.5 first enumerates all primitive shapes $A < B$, $N = A + B \le D_1$, satisfying

$$\mathrm{rad}(A)\,\mathrm{rad}(B)\,\mathrm{rad}(N) \le 4\,398\,937.$$

This enumeration is exhaustive without scanning every $A$. Order the three radicals as $\rho_1 \le \rho_2 \le \rho_3$. Then

$$\rho_1 \le \lfloor 4\,398\,937^{1/3} \rfloor = 163, \qquad \rho_2 \le \lfloor 4\,398\,937^{1/2} \rfloor = 2097.$$

The program generates every integer at most $D_1$ with radical at most these two bounds, assigns the two smallest radical roles in all three possible ways, and reconstructs the remaining member from $A + B = N$. A canonical role ordering removes duplicates. A separate brute-force test checks this logic on small boxes.

For every coprime scale pair $1 \le r \le s \le 139$, the pair program then applies, in order:

(1) the support compatibility of Corollary 11.5 outside $\{2, 3\}$, allowing a missing prime only when it divides the opposite scale;

(2) the two support caps (18.6), the range $D_0 \le D \le D_1$, and disjointness of the two scaled additive triples;

(3) the exact integer bounds $d > D/140$, $d^3 < 64D^2$, Theorem 14.2, and the proper-factor inequalities

$$sk \le N - 4, \qquad rk \le M - 4.$$

Here global isolation of an irreducible primitive base makes each selected primitive factor proper; Theorem 10.1 then rules out a degree-one complementary factor.

(4) the role-binomial bounds of Lemma 18.7;

(5) the full-radical congruences

$$sk \equiv 0, -2 \pmod{p} \quad (p \mid ABN), \qquad rk \equiv 0, -2 \pmod{p} \quad (p \mid CEM),$$

combined by an exact Chinese-remainder interval test.

No irreducibility oracle, floating-point cutoff, finite-field factorization, or modular gcd is used in this main pipeline. The deterministic output is:

| | |
|---|---|
| radical-admissible primitive shapes | 237 418 |
| scaled records | 2 450 291 |
| coprime scale pairs | 5 952 |
| shape pairs after range, support cap, and disjointness | 67 730 966 |
| after the degree interval | 2 124 295 |
| after role-binomial bounds | 139 640 |
| after full-radical CRT feasibility | 761 |
| same-shape rows among these | 761 |
| rows surviving Proposition 18.8 | 0. |

**Theorem 18.9** (Computer-assisted elimination of the unit range)**.** *There is no normalized counterexample with*

$$6\,816\,242 \leq \max\{m, n, p, q\} \leq 175\,394\,637.$$

*Consequently every normalized counterexample satisfies*

$$\boxed{\max\{m, n, p, q\} \leq 6\,816\,241.} \tag{18.10}$$

*Proof.* Every counterexample in the displayed range satisfies all of the mathematical filters just listed, so the exhaustive exact program places it among the 761 final rows. In every row the two primitive shapes agree. Proposition 18.8 gives

$$d \leq 2rs \leq 2 \cdot 139 \cdot 138 = 38\,364,$$

whereas Proposition 18.2 gives $d \geq 48\,688$. Thus all rows are impossible. Combining this with Theorem 17.13 proves (18.10). □

**Remark 18.10** (Certificate boundary)**.** The theorem is computer-assisted in the ordinary source-verification sense. The bundle contains the complete shape list, the 761 pre-final rows, an independent Python verifier for every recorded row, deterministic checksums, and a brute-force small-box regression test for the shape enumerator. The coverage of the large pair scan is encoded transparently in the short C++ loop rather than in a proof-assistant certificate. A future Lean-oriented version should replace that last trust boundary by interval or branch coverage certificates; no large polynomial arithmetic would be required.

## 18.7 Certified constants and reproduction commands

The archived upper-range verifier is replayed from this bundle by

```
make q12-verify-upper
```

The historical Qab9 upper generator can also be rerun from `archive/Qab9_upper_bundle` using its local Makefile. That pipeline requires a C++20 compiler with OpenMP and Python 3; its Python verification uses only the standard library. The exact rational certificate `archive/Qab9_upper_bundle/code/certify_constants.py` prints

```
plastic_power_coefficients=62608681,109870576,82938844
theta70_rational_lower=220094051941/625
twice_D_max=350789274
uniform_degree_bound=d>D/140
scale_max=139
same_shape_max=38364
unit_degree_integer_min=48688
support_endpoint_upper_floor=4398936
support_product_cap=4398937
```

The two analytic transition points in Theorem 17.13 and Corollary 17.14 are independently checked by the supplied Sage and Arb script `archive/Qab9_upper_bundle/code/certify_thresholds.sage`. All comparisons in both scripts are interval or exact rational comparisons, not binary floating-point guesses.

## 18.8 Exact factorization through total sixty

Exact factorization over $\mathbb{Q}$ was rerun with SymPy 1.14 for every unordered unequal pair with total $a + b \leq 60$. There are

$$\sum_{N=3}^{60} \left\lfloor \frac{N-1}{2} \right\rfloor = 870$$

such pairs. Every $Q_{a,b}$ was irreducible, and no normalized reciprocal factor orbit occurred in two different packages. The calculation took exact quotients and exact rational factorization; no numerical root matching was used.

For every tested pair, its primitive reduction also has total at most 60 and was found irreducible. Theorem 14.9 therefore upgrades all 870 finite factorizations to global certificates: every one of these packages is coprime to every distinct package in the entire infinite family, including packages of arbitrarily large total.

The complete script is included below.

```
from math import gcd
import sympy as sp
from sympy import Poly, QQ

x = sp.symbols("x")
MAXN = 60
seen, reducible, collisions = {}, [], []
count = globally_certified = 0

def qpoly(a, b):
    g = gcd(a, b)
    num = Poly(a*x**(a+b) - (a+b)*x**a + b, x, domain=QQ)
    den = Poly((x**g - 1)**2, x, domain=QQ)
    return num.exquo(den)

def reciprocal_orbit_key(f):
    p = Poly(f, x, domain=QQ).monic()
    rev = Poly.from_list(list(reversed(p.all_coeffs())),
                         gens=x, domain=QQ).monic()
    return min(tuple(p.all_coeffs()), tuple(rev.all_coeffs()))

for N in range(3, MAXN + 1):
    for a in range(1, (N - 1)//2 + 1):
```

```
        b = N - a
        count += 1
        q = qpoly(a, b)
        _, factors = sp.factor_list(q.as_expr(), x, domain=QQ)
        is_irreducible = (len(factors) == 1 and factors[0][1] == 1
                and Poly(factors[0][0], x).degree() == q.degree())
        if not is_irreducible:
            reducible.append((a, b))
        g = gcd(a, b)
        base = qpoly(a//g, b//g)
        _, base_factors = sp.factor_list(base.as_expr(), x, domain=QQ)
        base_irreducible = (len(base_factors) == 1
                and base_factors[0][1] == 1
                and Poly(base_factors[0][0], x).degree() == base.degree())
        if base_irreducible:
            globally_certified += 1
        for f, exponent in factors:
            key = reciprocal_orbit_key(f)
            if key in seen and seen[key] != (a, b):
                collisions.append((seen[key], (a, b)))
            else:
                seen[key] = (a, b)

print(count, reducible, collisions, globally_certified)
# Output: 870 [] [] 870
```

# 19 Lower-range extraction degrees and endpoint states

The upper computation of Section 18 gives

$$D := \max\{m, n, p, q\} \leq 6\,816\,241.$$

This section records the lower-range identities in a form suitable for exact enumeration, and treats in full the exceptional fourth-power case of Capelli's criterion. Write the two normalized packages as

$$\{ra, rb\}, \qquad \{sc, se\}, \qquad \gcd(a, b) = \gcd(c, e) = 1, \qquad \gcd(r, s) = 1,$$

and let $\alpha$ be a common root of selected ordered orientations. Put

$$F = \mathbb{Q}(\alpha), \quad K = \mathbb{Q}(\alpha^r), \quad K' = \mathbb{Q}(\alpha^s),$$

$$d = [F : \mathbb{Q}], \quad e_1 = [K : \mathbb{Q}], \quad e_2 = [K' : \mathbb{Q}], \quad t = [F : K], \quad u = [F : K'].$$

## 19.1 Extraction degrees and the reduced correspondence

**Lemma 19.1** (Extraction-degree identities)**.** *One has*

$$1 \leq t \leq r, \qquad 1 \leq u \leq s, \qquad d = te_1 = ue_2, \qquad d \leq e_1 e_2. \tag{19.1}$$

*Moreover $F = KK'$.*

*Proof.* The element $\alpha$ satisfies $X^r - \alpha^r$ over $K$, and similarly on the second side. The tower law gives the first identities. An integral Bézout combination of the coprime integers $r, s$ recovers $\alpha$ from $\alpha^r, \alpha^s$, so $F = KK'$. Finally $[KK' : \mathbb{Q}] \leq [K : \mathbb{Q}][K' : \mathbb{Q}]$. $\square$

**Proposition 19.2** (General reduced correspondence). *Suppose selected ordered primitive orientations $Q_{a,b}(x^r)$ and $Q_{c,e}(x^s)$ share a noncyclotomic irreducible factor of degree $d$. Put*

$$g_0 = \gcd(rb, se), \quad B_0 = \frac{rb}{g_0}, \quad E_0 = \frac{se}{g_0},$$
$$g_1 = \gcd(ra, sc), \quad A_0 = \frac{ra}{g_1}, \quad C_0 = \frac{sc}{g_1}.$$

*Unless the normalized packages are identical,*

$$\boxed{d \le E_0A_0 + B_0C_0 = \frac{rs(ae+bc)}{\gcd(rb, se)\gcd(ra, sc)}.} \tag{19.2}$$

*Proof.* Set $x = \alpha^r$, $y = \alpha^s$, $N = a + b$, $M = c + e$, and

$$U_0 = \frac{a}{N}x^b, \qquad 1 - U_0 = \frac{b}{N}x^{-a}, \qquad V_0 = \frac{c}{M}y^e, \qquad 1 - V_0 = \frac{e}{M}y^{-c}.$$

The identity $x^s = y^r$ gives two curves in $\mathbb{P}^1 \times \mathbb{P}^1$:

$$U_0^{E_0} = \lambda_0 V_0^{B_0}, \qquad (1 - U_0)^{C_0} = \lambda_1(1 - V_0)^{A_0}.$$

Their bidegrees are $(E_0, B_0)$ and $(C_0, A_0)$, hence their intersection number is $E_0A_0 + B_0C_0$. A common curve component would, by Lemma 17.2, make $V_0$ one of the six Möbius transforms of $U_0$. The divisor table used in Proposition 18.8 then leaves only $V_0 = U_0$, $se = rb$, and $sc = ra$, which is the identical-package case.

The value $U_0$ determines both $x^b$ and $x^{-a}$, and therefore $x$, since $\gcd(a, b) = 1$; similarly $V_0$ determines $y$. The pair determines $\alpha$ because $\gcd(r, s) = 1$. Thus the $d$ conjugates give $d$ distinct intersection points, and Bézout proves the bound. □

For equal *selected ordered* orientations, (19.2) gives $d \le 2rs$. When the unordered primitive shapes agree but the selected orientations are opposite, Proposition 18.8 still gives the stronger bound $d \le 2rs$; the two statements should not be conflated.

## 19.2 Norm transfer, endpoint arithmetic, and packet occupancy

Let

$$h_\alpha(X) = UX^d + \cdots + V \in \mathbb{Z}[X], \qquad U > 0,$$

be the primitive minimal polynomial of $\alpha$, and let the primitive minimal polynomials of $\beta = \alpha^r$ and $\gamma = \alpha^s$ have leading and constant coefficients $(u_1, v_1)$ and $(u_2, v_2)$.

**Lemma 19.3** (Norm transfer). *One has*

$$u_1^t = U^r, \qquad |v_1|^t = |V|^r, \qquad u_2^u = U^s, \qquad |v_2|^u = |V|^s. \tag{19.3}$$

*Proof.* Gauss's lemma applies because the relevant primitive factor divides a primitive composition of $Q_{a,b}$, whose leading and constant coefficients divide the coprime integers $a, b$. Hence the leading and constant coefficients of each primitive irreducible factor are coprime; in particular $V/U$ and $v_1/u_1$ are reduced fractions. The norm identity

$$N_{F/\mathbb{Q}}(\alpha^r) = N_{K/\mathbb{Q}}(\beta)^t$$

then reads $(V/U)^r = (v_1/u_1)^t$, up to sign. Comparing numerator and denominator valuations gives the first pair of identities; the second is identical. □

**Proposition 19.4** (Scaled endpoint divisibilities and slopes)**.** *For the first selected orientation,*

$$|V|^{ra} \mid b^d, \qquad U^{rb} \mid a^d; \tag{19.4}$$

*for the second,*

$$|V|^{sc} \mid e^d, \qquad U^{se} \mid c^d. \tag{19.5}$$

*If $p \mid U$, then*

$$\frac{v_p(a)}{rb} = \frac{v_p(c)}{se}; \tag{19.6}$$

*and if $q \mid V$, then*

$$\frac{v_q(b)}{ra} = \frac{v_q(e)}{sc}. \tag{19.7}$$

*In particular $U \mid a, c$ and $|V| \mid b, e$.*

*Proof.* Apply Proposition 17.20 to the primitive factor of $Q_{a,b}$, raise its divisibilities to the extraction degree $t$, and use Lemma 19.3 together with $te_1 = d$. The slope identities follow by choosing embeddings with negative, respectively positive, valuation and reading the endpoint Newton slopes. The final assertion also follows directly from Gauss's lemma. □

**Proposition 19.5** (Exact endpoint packet occupancy)**.** *Suppose $p^\kappa \| a$, $p \mid U$, and put*

$$\lambda = \frac{r}{t} v_p(U) \in \mathbb{Z}_{>0}.$$

*If the primitive factor of $Q_{a,b}$ has degree $e_1$, then its number of roots in the negative-valuation endpoint block is*

$$w_p = \frac{b\lambda}{\kappa} \in \mathbb{Z}, \qquad 1 \le w_p \le \min\{b, e_1\}, \qquad e_1 - w_p \equiv 0 \text{ or } -2 \pmod{p}. \tag{19.8}$$

*There is a reciprocal statement for $p^\kappa \| b$ and $p \mid V$.*

*Proof.* Every root in the nonunit endpoint block has valuation $-\kappa/b$. The factor leading coefficient has valuation $v_p(u_1) = rv_p(U)/t = \lambda$, while its constant coefficient is a $p$-adic unit. The product-of-roots formula therefore gives $w_p\kappa/b = \lambda$. The remaining roots lie in the unit packets; the center-1 packet contributes either zero or $p-2$ roots and all other clusters contribute multiples of $p$, proving the congruence. □

### 19.3 Sharper coefficient cutoffs

**Proposition 19.6** (Two nonunit coefficients)**.** *If $U > 1$ and $|V| > 1$, then*

$$\boxed{D \le 16\,583.} \tag{19.9}$$

*Proof.* Choose distinct primes $p \mid U$, $q \mid V$. Equations (19.6)–(19.7) and Proposition 19.2 give

$$d \le v_p(c)v_q(b) + v_p(a)v_q(e) \le 2\lfloor \log_2 D \rfloor \lfloor \log_3 D \rfloor.$$

The plastic number $\theta_0$ exceeds $1324717/10^6$, as follows by substituting this rational number into $X^3 - X - 1$. The height estimate therefore gives the strict lower bound

$$d > \frac{D \log(1324717/10^6)}{2 \log(2D)}.$$

On each interval on which the two logarithmic floors are constant, the right side is increasing. Exact Arb interval comparisons at the finitely many power-of-two and power-of-three boundaries show that the two bounds are incompatible for every $D \geq 16\,584$; the comparison is certified by the analytic-constants verifier listed in Table 1. □

**Corollary 19.7** (Coefficient trichotomy). *For $16\,584 \leq D \leq 6\,816\,241$, at most one of $U, |V|$ is a nonunit and*

$$\max\{U, |V|\} \leq 26.$$

*For $D \geq 50\,000$, the upper bound improves to 12.*

*Proof.* The function $8\log(2D)/D^{1/3}$ is decreasing in this range. The Arb certificate proves that its value at 16 584 is below $\log 27$, and its value at 50 000 is below $\log 13$. The absolute leading and constant coefficients of a primitive polynomial are at most its Mahler measure. Proposition 19.6 excludes two nonunits. □

### 19.4 The exceptional fourth-power branch

**Lemma 19.8** (Capelli's exceptional branch). *Let $K = \mathbb{Q}(\beta)$, and suppose*

$$\beta = -4\eta^4 \qquad (\eta \in K).$$

*Then the exceptional branch cannot occur in a total-role packet or in a 2-adic unit endpoint packet. If it occurs in the leading nonunit endpoint block of $Q_{a,b}$, where $2^\kappa \| a$, then*

$$e_1 \leq b, \qquad 2b \mid e_1\kappa. \tag{19.10}$$

*The reciprocal assertion holds for the constant endpoint.*

*Proof.* The element $-\beta = (2\eta^2)^2$ is a square in $K$. In every 2-adic unit packet used in the local power analysis, after an unramified extension there is an odd-order root of unity $\zeta$ and an integer $\ell > 1$ such that

$$v_2(\beta - \zeta) = 1/\ell < 1.$$

Since $v_2(2\zeta) = 1$, the ultrametric inequality gives

$$v_2(-\beta - \zeta) = v_2((\beta - \zeta) + 2\zeta) = 1/\ell.$$

The Newton-edge power obstruction, applied to $-\beta$, now contradicts its being a square. This excludes total packets and the unit part of an endpoint packet.

Every conjugate of $\beta$ is again of the form $-4\eta_i^4$. Hence a surviving irreducible factor has all its roots in the nonunit endpoint block, which has length $b$; thus $e_1 \leq b$. Finally

$$N_{K/\mathbb{Q}}(\beta) = (-4)^{e_1} N_{K/\mathbb{Q}}(\eta)^4$$

has even 2-adic valuation, while the endpoint slope gives $|v_2 N_{K/\mathbb{Q}}(\beta)| = e_1\kappa/b$. Therefore $2b \mid e_1\kappa$. □

**Proposition 19.9** (Localization of a Kummer defect). *Let $p^A \| r$. If $v_p(t) < A$, then either $\beta \in K^p$, or $p = 2$, $A \geq 2$, and $\beta \in -4K^4$. The following alternatives exhaust the branch:*

*(a) if $p$ is good for the primitive shape, then $e_1 \leq p - 2$;*

*(b) a total-role prime is impossible;*

*(c) a locally $p$-adic unit endpoint factor is impossible;*

*(d) in a nonunit endpoint block,*

$$e_1 \le b, \qquad pb \mid e_1\kappa$$

*for a leading endpoint $p^\kappa \| a$, with the reciprocal statement at the constant endpoint. Necessarily $p \in \{2, 3, 5, 7\}$.*

*In particular every unit-coefficient Kummer-defect counterexample satisfies*

$$D < 140 \cdot 139 \cdot 137 = 2\,666\,020, \qquad d \le 139 \cdot 137 = 19\,043. \tag{19.11}$$

*Proof.* Capelli's criterion gives the first dichotomy. In the ordinary internal $p$-th-power branch, Theorem 14.5, Corollary 15.5, and Theorem 15.6 give (a)–(d). Lemma 19.8 supplies exactly the missing exceptional $p = 2$ case. The endpoint condition implies $\kappa \ge p$; since $p^\kappa \le 6\,816\,241$, only $2, 3, 5, 7$ remain. For a unit factor every deficient prime is therefore good, so $e_1 \le p - 2 \le 137$, and (19.11) follows from $d = te_1 \le re_1$, $r \le 139$, and $d > D/140$. □

### 19.5 A strict endpoint-state object

**Definition 19.10** (Admissible lower state)**.** A package state records

$$(r, t, e_1, d, U, V; a, b, \varepsilon)$$

together with every deficient scale prime, its valuations in $r, t$, its role, and its endpoint exponent $\kappa$. A pair of package states is *admissible* if it has common $(d, U, V)$, satisfies

$$d = te_1 = ue_2, \qquad d \le e_1e_2, \qquad 140d > D, \qquad d^3 < 64D^2, \qquad d \le \min\{a + b, c + e\} - 2, \tag{19.12}$$

all norm-transfer, endpoint, slope, packet-occupancy, radical-signature, and correspondence conditions above, the coefficient trichotomy of Corollary 19.7, the alternatives of Proposition 19.9, coprime scales, and disjoint scaled additive triples.

In the verification bundle these fields are represented by typed CSV rows. The separate verifiers listed in Table 1 recompute the global integer conditions from those rows, and the small-box generators supply coverage regressions for the enumeration logic.

**Theorem 19.11** (Finite lower-state envelope)**.** *Every normalized counterexample with $D \le 6\,816\,241$ determines an admissible lower state. The state set is finite and effectively enumerable.*

*Proof.* The preceding lemmas and the exact local factor signatures give all listed necessary conditions. The explicit bounds on $D, r, s, d, U, V$, together with the endpoint divisibilities, leave only finitely many integer tuples. □

## 20 Exact elimination of the complete unit-coefficient branch

After the upper range is gone, a normalized counterexample in the lower box has either unit endpoint coefficients or at least one nonunit endpoint coefficient. This section completes the unit branch by combining full-Kummer searches, modular gcd certificates, and modular factor-degree certificates.

This section covers the whole remaining box under the hypothesis

$$U = |V| = 1,$$

including both full-Kummer and Kummer-defect cases. All branch decisions in the C++ enumerators are integer decisions. The one analytic post-filter uses Arb balls and rounds in the conservative direction.

**Lemma 20.1** (Subset-degree irreducibility certificate)**.** *Let $F \in \mathbb{Z}[x]$ be primitive of degree $n$, and let $p \nmid \mathrm{lc}(F)$. If the reduction of $F$ modulo $p$ has irreducible factor degrees $d_{p,j}$, counted with multiplicity, then the degree of every factor of $F$ over $\mathbb{Q}$ is a subset sum of the $d_{p,j}$. Consequently, if the intersections of these subset-sum sets for finitely many primes is $\{0, n\}$, then $F$ is irreducible.*

*Proof.* A primitive rational factor may be chosen in $\mathbb{Z}[x]$. Since the total leading coefficient is nonzero modulo $p$, neither factor loses degree on reduction. Its reduction is a divisor of $F \bmod p$, so its degree is the sum of a submultiset of the modular irreducible factor degrees. □

**Theorem 20.2** (Unit-coefficient elimination)**.** *There is no normalized counterexample whose primitive minimal polynomial has leading and constant coefficients of absolute value one.*

*Computer-assisted proof.* The calculation has three exhaustive branches. In each branch the search is conservative: a shape or pair is retained whenever a local theorem is not being used as a certified filter, and it is discarded only by an independently recomputed arithmetic certificate.

**(I) Full Kummer degree and $50\,000 \leq D \leq 6\,816\,241$.** The one-package support estimate is bounded throughout this interval by

$$r \ \mathrm{rad}(ab(a+b)) < 1\,612\,000.$$

This is the same envelope $C(D)$ as in Corollary 18.5, but restricted to $D \geq 50\,000$; the analytic-constants verifier listed in Table 1 prints the exact rational certificate

$$C(50\,000) < 1\,612\,000.$$

The generator keeps 63 980 primitive shapes. The scale-pair scan considers 334 910 032 indexed shape pairs and applies only the exact radical, height, role-binomial, modular factor-degree, and correspondence filters. The conservative terminal list consists of seven package pairs. For each of the four selected orientation pairs attached to each package pair, the certificate records a good auxiliary prime at which the two collision trinomials have modular gcd of degree exactly two. The independent verifier reconstructs all twenty-eight modular computations.

**(II) Full Kummer degree and $D < 50\,000$.** Here $d \leq 5428$. For every primitive factor degree $e$, the relation $\mathrm{rad}(ab(a+b)) \mid e(e+2)$ permits direct smooth-number enumeration. It produces 24 301 shape–degree package records. The exact pair scan reduces 734 084 raw state pairs to four terminal state pairs and three terminal package pairs. Their twelve selected orientation pairs are certified by the same modular-gcd criterion, and the verifier recomputes every claimed gcd degree.

**(III) A deficient scale prime.** Proposition 19.9 gives $D < 2\,666\,020$, $d \le 19\,043$, and a deficient-side primitive degree at most 137. Enumerating extraction cores first and then the smooth additive triples reduces the branch to 21 114 integer state pairs, representing 9297 package-orientation pairs but only 46 primitive shapes.

Lemma 20.1 certifies every one of these 46 shapes as irreducible. The largest shape $(1, 6655)$ is verified with several auxiliary primes; the certificate file stores the exact modular factor-degree lists and the separate verifier recomputes 106 modular factorizations. The global isolation theorem for irreducible primitive bases then excludes every row in this branch.

The branch generators, output tables, certificates, and independent recomputers are listed in Table 1. The bundled checksums and the unit-branch manifest provide the row counts, terminal counts, and SHA-256 bindings used for replay. □

| branch | principal input | preterminal rows | terminal objects |
|---|---|---|---|
| full Kummer, $D \ge 50\,000$ | 101 752 371 pairs | 7 pairs | 28 gcd certificates |
| full Kummer, $D < 50\,000$ | 7 366 972 pairs | 3 pairs | 12 gcd certificates |
| Kummer defect | state-pair scan | 46 shapes | 46 irreducibility certificates |

Table 2: The three unit-coefficient branches. Counts are recomputed from the bundled deterministic artifacts; timing data are kept outside the proof manifest.

**Corollary 20.3** (Remaining nonunit envelope)**.** *Every normalized counterexample has $U > 1$ or $|V| > 1$. After replacing $\alpha$ by $\alpha^{-1}$ and reversing both selected orientations when necessary, the one-nonunit branch may be written with $U > 1$, $|V| = 1$. Moreover:*

*(i) if both coefficients are nonunits, then $D \le 16\,583$;*

*(ii) if $D \ge 16\,584$, then $2 \le U \le 26$;*

*(iii) if $D \ge 50\,000$, then $2 \le U \le 12$;*

*(iv) every deficient endpoint prime lies in $\{2, 3, 5, 7\}$ and satisfies (19.8), (19.10), and the slope identities.*

## 21 Exact nonunit branch artifacts

The final branch keeps the endpoint coefficient data instead of collapsing to full Kummer degree. The state records here are designed so that the verifiers can recompute norm transfer, endpoint occupancy, slope signatures, and terminal modular gcd certificates from explicit rows.

This section presents the nonunit computation. It is kept separate from the unit proof because the nonunit package states retain exact extraction degrees and endpoint packet occupancies rather than full Kummer multiplicity.

**Theorem 21.1** (Both-nonunit elimination)**.** *There is no normalized counterexample for which both the leading coefficient and the absolute constant coefficient of the primitive minimal polynomial are nonunits.*

*Computer-assisted proof.* Proposition 19.6 and the strengthened constants of Corollary 20.3 reduce this branch to $D \le 16\,583$, $d \le 224$, and endpoint valuations at most 14. The generator enumerates every one-package state satisfying the exact norm transfer, endpoint divisibility, slope-signature, packet-occupancy, deficient-prime, total-radical, degree, and size conditions. It finds 1054 package states. The independent verifier recomputes every row and checks that the common-factor keys

$$(U, V, d; \sigma_U, \sigma_V)$$

are all distinct. Hence two packages cannot share the same factor data; the state-pair file is empty, as confirmed by the independent verifier listed in Table 1. $\square$

**Theorem 21.2** (One-nonunit elimination above 16 583)**.** *There is no normalized counterexample with exactly one nonunit endpoint coefficient and $D \geq 16\,584$.*

*Computer-assisted proof.* After reciprocal normalization write $U > 1$, $|V| = 1$. The exact coefficient-transition certificate gives $2 \leq U \leq 26$. For each such $U$, the bound

$$\log U < \frac{8\log(2D)}{D^{1/3}}$$

gives the sharp coefficient-specific cap recorded by the analytic-constants verifier listed in Table 1: the largest cap is 6 816 241 for $U = 2$, and the cap for $U = 26$ is 16 743. The same certificate supplies an integer $C(U)$ such that $D < C(U)d$ throughout the corresponding coefficient range.

The all-shapes generator is conservative: it keeps primitive shapes even when a known irreducibility theorem would already discard them. Across $U = 2, \ldots, 26$ it produces 608 490 package records, grouped into 514 280 common-factor keys. The pair scan tests 137 530 raw same-key package pairs. The successive exact filters leave

| | |
|---:|---|
| 10 636 | after coprime scales, |
| 7 916 | after range and extraction-degree checks, |
| 5 430 | after disjointness, |
| 483 | after support compatibility, |
| 76 | after the global degree bounds, |
| 0 | after the local role-binomial bound. |

Thus no terminal modular gcd computation is needed in this branch. The independent verifier listed in Table 1 recomputes the row conditions, confirms the per-$U$ pair counts, and verifies that every survivor file is header-only. $\square$

**Corollary 21.3** (Residual envelope before the final small computation)**.** *If a normalized counterexample remains after the computations in this bundle, then after replacing $\alpha$ by $\alpha^{-1}$ and reversing both selected orientations if necessary, its primitive minimal polynomial has*

$$U > 1, \qquad |V| = 1, \qquad D \leq 16\,583.$$

*Moreover $U \mid a$, hence $2 \leq U \leq D$, every endpoint valuation is at most 14, the safe degree bound*

$$d^3 < 64D^2$$

*gives $d \leq 2601$, and every deficient endpoint prime that lies in the nonunit endpoint block lies in $\{2, 3, 5, 7\}$.*

**Theorem 21.4** (Residual one-nonunit elimination)**.** *There is no normalized counterexample with exactly one nonunit endpoint coefficient and $D \leq 16\,583$.*

*Computer-assisted proof.* By Corollary 21.3, after reciprocal normalization it is enough to enumerate

$$D \le 16\,583, \qquad |V| = 1, \qquad 2 \le U \le D, \qquad d \le 2601.$$

The residual generator uses five slope-signature slots, enough because an integer at most 16 583 has at most five distinct prime factors. It enumerates every one-package state satisfying the norm-transfer exponent integrality, endpoint divisibility, packet occupancy, slope congruence, deficient-prime, total-radical, extraction-degree, and size conditions. It is conservative and keeps rows that later degree filters can remove. The output is

| | |
|---|---|
| 233 278 | package records, |
| 212 369 | common-factor keys, |
| 29 611 | raw same-key state pairs, |
| 5 680 | after coprime scales and the residual range, |
| 5 679 | after extraction-degree checks, |
| 2 345 | after disjointness, |
| 428 | after support compatibility, |
| 234 | after the global degree bounds, |
| 4 | after the local role-binomial bound, |
| 4 | after the reduced correspondence bound. |

The four terminal state pairs collapse to two package pairs:

$$\begin{aligned} U &= 8, & d &= 99: & &(3; 440, 1, 441) \text{ vs. } (4; 3024, 1, 3025), \\ U &= 32, & d &= 123, 205, 285: & &(1; 6560, 1, 6561) \text{ vs. } (2; 1024, 1, 1025). \end{aligned}$$

For each of these two package pairs and each of the four selected orientations, the recorded residual certificates give the prime $p = 1009$ and modular gcd degree 2. The independent residual verifier listed in Table 1 recomputes every package row, reruns the pair sieve, compares the terminal CSV files exactly, and recomputes all eight modular gcds over $\mathbf{F}_{1009}$. Degree 2 is accounted for by the forced double root at $x = 1$, so no noncyclotomic common factor remains in any terminal orientation. □

### Assurance status of the verification bundle

The bundled verification artifacts eliminate the upper range, the full unit branch, the both-nonunit branch, the one-nonunit branch with $D \ge 16\,584$, and the residual one-nonunit branch with $D \le 16\,583$. The light verifier replays the nonunit row/count checks, the residual pair sieve, the residual modular gcd certificates, and the small brute-force oracle for the residual generator.

## Conclusion

The mathematical reduction and the upper computation first give

$$D \le 6\,816\,241.$$

The exceptional $-4K^4$ branch is handled separately, so deficient scale primes are localized without citing an ordinary internal-power theorem outside its hypotheses. The exact unit computation eliminates every unit-coefficient common factor. The nonunit artifacts eliminate both endpoint coefficients being nonunit and also eliminate the whole one-nonunit branch above 16 583.

The residual one-nonunit computation then eliminates the remaining range $D \leq 16\,583$, $2 \leq U \leq D$, $d \leq 2601$, using four terminal state pairs and eight modular gcd certificates.

Consequently no normalized reciprocal-package counterexample remains in the finite box reached by the mathematical reductions. By the collision dictionary and primitive normalization of Section 2, this proves Theorem 1.1. The present bundle is therefore a reviewable computer-assisted proof package for the reciprocal-package coprimality statement.

# A Computer-Assisted Proof Contract

This appendix records the verification contract behind the finite computations. The scripts are deterministic and use exact integer or finite-field arithmetic. A generator is trusted only for exhaustive coverage of the rows it emits; the corresponding verifier checks the row schema, recomputes all stated integer predicates or terminal certificates listed below, and compares bundled hashes or manifests.

**Upper range.**
The archived Qab9 pipeline enumerates all rows in the certified range. `verify_upper_safe.py` checks 237,418 shapes, 761 prefinal pairs, and no residual candidates. The remaining trust is archived coverage.

**One-nonunit $D \geq 16\,584$.**
The per-$U$ package generators enumerate all admissible rows for $U = 2, \ldots, 26$. `verify_one_nonunit.py` checks schemas, row predicates, counts, and empty terminal CSVs. The remaining trust is generator coverage.

**Residual one-nonunit.**
The residual generator enumerates the corrected $d \leq 2601$ range. `verify_one_nonunit_residual.py` reruns the pair sieve, compares terminal CSVs, and recomputes eight gcd certificates over $\mathbf{F}_{1009}$. The remaining trust is generator coverage.

**Unit full-Kummer branches.**
The unit generators enumerate the large and small unit package rows. `verify_unit_branch_manifest.py` checks stage counts, hashes, terminal pair counts, and certificate coverage; `verify_modular_gcd_certificates.py` recomputes the FLINT gcds. The remaining trust is unit enumeration coverage.

**Unit deficient-scale branch.**
The defect generators enumerate the Kummer-defect rows. The unit manifest checks the row and terminal counts, and `verify_irreducibility_certificates_direct.py` recomputes the 46 irreducibility certificates. The remaining trust is defect enumeration coverage.

The principal CSV schemas are:

```
residual packages:
U,d,sigma1,...,sigma5,r,t,e,a,b,n

residual state pairs:
U,d,sigma1,...,sigma5,r,t,e1,a,b,n,s,u,e2,c,f,m,D,correspondence_bound,
   role_bound
```

```
residual terminal orientations:
U,r,a,b,n,s,c,f,m

unit full-Kummer terminal orientations:
r,a,b,n,s,c,f,m

unit modular certificate extension:
pair_index,orientation,low1,low2,prime,gcd_degree

unit-defect state pairs:
d,r,t,e1,a,b,n,defect1,s,u,e2,c,f,m,defect2,D,role_bound,
   correspondence_bound
```

The public non-mutating verification command is `make verify-light`. The FLINT-dependent arithmetic replay is `make verify-unit`. The residual manifest is checked with `-check-manifest`; it is regenerated only with the explicit `-write-manifest` mode.

# B Local Packet Valuation Table

The local packet arguments use the following cases, all with the conventions translated from Borisov's notation in Sections 11 and 14. For a prime $p$ dividing exactly one role of a primitive triple, Borisov's endpoint lemmas give the packet center and slope denominator. At a total prime $p \mid A + B$, every unit root lies in a nontrivial root-of-unity packet, and the packet contribution is divisible by $p$ except for the center-1 residue already recorded in the role congruence. At a leading endpoint prime $p \mid A$, Lemma 2.10 of [1] gives the nonunit block of length $B$ and the unit packet counted by Lemma 2.11; the constant endpoint is reciprocal. If $p = 2$, the first two formal layers must be kept merged, producing the residual split used in Lemma 15.3. This is the only place where Capelli's exceptional $-4K^4$ branch can survive, and Lemma 19.8 shows that a survivor is forced into the nonunit endpoint block with $e \leq B$ and the stated norm divisibility.

# Use of AI tools

This project used AI tools in discovery, editing, and implementation. The initial proof strategy was found through multiple rounds of prompting to ChatGPT 5.5 Pro. Claude Code (Opus 4.8 max) was used for manuscript-level revision and audit-oriented cleanup. The Codex CLI tool running gpt-5.5 xhigh edited the manuscript, implemented and revised verification scripts, finalized the computer-verification artifacts, and ran the reproducibility checks. The mathematical claims in the paper are supported by the written arguments and by the deterministic proof artifacts described in Appendix A; the AI transcripts themselves are not used as proof evidence.

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